Pooling and Drift in Delayed Bandits

Melika Baghi
Georgia Institute of Technology
mbaghi3@gatech.edu

Abstract

An action's consequences often arrive in two stages: an intermediate signal at once, and the outcome that matters only after a delay, as a recommender sees a click in seconds but a purchase in days. Over T rounds with K actions and delay d, the rate available when the state-to-outcome map is unrestricted is $\widetilde{O}(\sqrt{(K+d)T})$ (Esposito et al., 2023), with $\widetilde{O}$ hiding logarithmic factors, so costs grow with the catalogue even when many items behave alike. When can the intermediate signal remove that dependence? If the outcome mean depends on the action only through the state it produced, one delayed outcome informs every action that could have produced it, and the cost is set not by how many actions there are but by how differently they behave, an *effective dimension* v_t between 1 and the number of states $|\mathcal{S}|$. Running $d+1$ copies of exponential weights in rotation attains $\widetilde{O}(\sqrt{(d+1)V\log K})$ for any $V \geq \sum_t v_t$ fixed in advance; a single copy makes the delay additive instead. Grouping similar states lowers the cost further, at a bias we make explicit. Sharing cannot remove the cost of delay itself: a learner handed the exact losses from d rounds ago still suffers $\Omega(\sqrt{d\mathcal{E}\min\{1+\log J, T/d\}})$, where J counts the drifting directions and $\mathcal{E}$ bounds how far the losses drift while it waits. Intermediate feedback therefore helps only when the signal is both shared across actions and still current. Synthetic experiments on the single-copy variant cut regret by up to 79 per cent against action-level weighting, and by 32 to 68 per cent against the tuned minimax-optimal method of Zimmert and Seldin (2020).

1 Introduction

Delayed outcomes are common in sequential decision-making: a system must act now even though the outcome it cares about may arrive much later. In recommendation, an item is shown immediately, an engagement signal such as a click or browsing depth is observed soon after, and conversion or retention may only be known days or weeks later, so decisions are committed before their economically relevant consequences appear. The central question is when such a signal can reduce the cost of learning before the final outcome arrives.

Here actions are catalogue items, states are intermediate engagement outcomes, and the delayed outcome is conversion. Different items may induce similar distributions over engagement states, so an outcome observed after one item can carry information about many others. The difficulty scales not with the catalogue size K but with how differently the actions distribute users across the intermediate states, so thousands of items funnelling users through a few engagement patterns stay easy to learn from.

Bandits with intermediate observations formalize this: the learner repeatedly picks one action and sees the consequence of that action alone. Esposito et al. (2023) show that the *state-to-loss* map decides the complexity, an unrestricted one giving only the rate available with no intermediate observations; we fix that map and ask how much the *action-to-state* structure helps.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

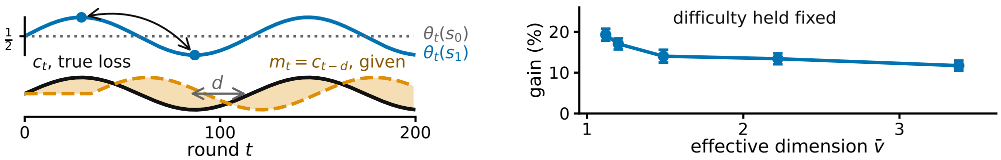

Figure 1: **Both panels synthetic. Left, the state-to-outcome relationship changes while the action-to-state channel stays fixed**, so the loss vector from d rounds earlier grows increasingly inaccurate. The drift measure E_2 of Section 3 tracks it. **Right, more overlap allows more sharing across actions**, at matched task difficulty, $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$, 20 paired seeds, bars one standard error of the paired difference. **The right panel runs the practical single-copy variant, which Theorem 1 does not cover and Theorem 2 does** (Appendix I, study 10).

Intermediate observations face two independent limitations: whether information can be shared *across actions*, and whether it stays useful *across time*.

Across actions: because similar actions reach similar states, we can *pool through the state*, charging a delayed outcome to every action that could have produced the state observed. The resulting statistical cost is governed by an *effective dimension* v_t, which measures how distinguishable the action-induced state distributions are under the learner's current play. When many actions behave similarly through the state channel, v_t can be far smaller than K.

Across time: delay creates a second difficulty even when the learner is handed an exact description of the past. Let $m_t = c_{t-d}$ be the action-loss vector from d rounds earlier. Its usefulness depends on how far the losses have moved, which we measure by $E_2 = \sum_t \|c_t - m_t\|_\infty^2$ (Figure 1). Our lower bound asks how much regret this mismatch forces even when m_t is supplied for free.

In short: with P known, running $d{+}1$ copies lets the states cut the cost of learning (Theorem 1), a single copy makes the delay additive (Theorem 2), and a grouping fixed in advance trades dimension against bias (Theorem 3); but no algorithm escapes the cost of delay, even one handed the stale losses outright (Theorem 4). The first three address the difficulty across actions and the last the one across time. An unknown action-to-state matrix can be estimated, with a weaker guarantee (Appendix E), while jointly adapting to overlap and temporal drift remains open (Conjecture 5).

2 Related work

Our results connect two largely separate lines of work: delayed feedback, and information sharing across actions. *Delayed bandits.* Vernade et al. (2020) vary the action-to-state map while fixing the state-to-loss map; we do the reverse, letting the latter vary obliviously. Zimmert and Seldin (2020) give the optimal $\widetilde{O}(\sqrt{KT} + \sqrt{D \log K})$ for total delay D, and Esposito et al. (2023) allow an unknown, possibly adversarial action-to-state map; fixing that channel is what makes its overlap geometry measurable through v_t. *Stale predictions.* Bar-On and Mansour (2025) derive upper bounds adapting to the inaccuracy of evolving observations; our lower bound instead asks what delay-dependent cost is unavoidable in stale-prediction error. Wang et al. (2026) bound squared gradient prediction error without delay, so the delay factor and its saturation are new here; Zhang et al. (2026) share the motivation in a stochastic Bayesian model where ours is adversarial. *Mediator feedback* is the closest structural connection: $v_t - 1$ is the χ^2 mutual information of Eldowa et al. (2024). Our contribution is not that quantity but its role when the outcome at the observed state is delayed, and the separation of that benefit from an independent temporal obstruction (Appendix F). Feedback graphs and off-policy coverage (Gabbianelli et al., 2023) are related forms of information sharing; Appendix A compares them.

3 Problem formulation

There are T rounds, K actions, a finite state space $\mathcal{S}$, a known action-to-state matrix P, and a fixed known delay d; all logarithms are natural and Appendix B tabulates every symbol. Every round follows the chain $A_t \xrightarrow{P} S_t \xrightarrow{\theta_t} X_t$: the learner draws A_t from a distribution x_t, observes

$S_t \sim P(\cdot \mid A_t)$ at once, and receives the delayed outcome only at time $t+d$, encoded as a loss $X_t \in [0,1]$ so that smaller is better, with $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$, where $\mathcal{F}_{t-1}$ is everything seen before round t, and outcomes conditionally independent across rounds given the states. The conditional mean depends on A_t only through S_t. This *state-sufficiency* assumption licenses cross-action pooling: an outcome observed at state S_t may update every action that could have produced that state. If an action affects the outcome directly, in a way not represented in S_t, the argument no longer applies. Here x_t depends only on $\mathcal{F}_{t-1}$; X_r may first affect the action distribution at round $r+d+1$, equivalently the round-t decision uses outcomes only from rounds $r \le t-d-1$.

That matrix $P \in [0,1]^{K\times|\mathcal{S}|}$ is row-stochastic and fixed, and the *state-loss means* $\theta_t \in [0,1]^{|\mathcal{S}|}$ are *oblivious*: chosen in advance, not reacting to the learner. Action losses are $c_t = P\theta_t$ and performance is the static *pseudo-regret*, the gap to the best single action in hindsight, $\mathbb{E}[R_T] = \mathbb{E}[\sum_t c_t(A_t)] - \min_a \sum_t c_t(a)$. **Oracle for the lower bound.** For Theorem 4 only, the learner also receives the *stale action-loss vector* $m_t \triangleq c_{t-d}$ before choosing A_t, with $m_t = c_1$ for $t \le d$. This is not a function of the observables, so a lower bound surviving it is stronger, and STATE-EXP3 never uses it. We measure how misleading m_t is by $E_2 = \sum_t \|c_t - m_t\|_\infty^2 \le T$, which stays small when errors are small even if frequent. The geometry of P thus governs sharing across actions and the evolution of θ_t governs staleness across time, the two axes of Sections 4 and 6.

4 Pooling through the state

When an outcome arrives, do not update only the action played: update every action by how likely it was to produce the observed state. Here P is known and the learner does not receive m_t. Let $q_t(s) = \sum_a x_t(a)P(s \mid a)$ be the chance of seeing state s this round, and give each action $\hat{c}_t(a) = P(S_t \mid a)X_t/q_t(S_t)$. It is correct on average, and how noisy it is depends on how much the actions overlap, through the *effective dimension* $v_t = \sum_s q_t(s)^{-1}\sum_a x_t(a)P(s \mid a)^2 \in [1, |\mathcal{S}|]$. STATE-EXP3 runs m copies of EXP3 (Auer et al., 2002) in rotation at learning rate η, copy i taking every mth round, and adds $\hat{c}_t$ to a copy's running total once the outcome is usable. A round costs $O(K)$ (Appendix C, with pseudocode and the χ^2 reading of v_t).

Theorem 1. *Let P be known and (θ_t) oblivious. Pick before the run any fixed number V with $\sum_t v_t \le V$ always. Then* STATE-EXP3 *with $m = d+1$ and the matching learning rate satisfies*

$$\mathbb{E}[R_T] \le \sqrt{2(d+1)V\log K}.$$

Writing $\bar{v}$ for the largest v_t over all ways of playing, $V = \bar{v}T \le |\mathcal{S}|T$ is one such choice, and it depends on P alone.

So K has been replaced by V, which counts how different the actions really are. The proof uses two facts, that the shared estimate is correct on average and no noisier than v_t, and that with $d+1$ copies in rotation each copy has its own previous outcome before it acts again. With $V = \bar{v}T$ this beats the general rate $\widetilde{O}(\sqrt{(K+d)T})$ (Esposito et al., 2023) whenever $\bar{v}(d+1)\log K \lesssim K+d$. Three gaps remain: rotating copies multiplies the delay cost by $d+1$; the bound uses v_t where $v_t - 1$ would be correct, so it stays at $\sqrt{2(d+1)T\log K}$ even when every action behaves alike and the true regret is zero; and it ignores E_2. The first is an artifact of the rotation, and dropping it costs nothing.

Theorem 2. *Under the hypotheses of Theorem 1, pick any $V^- \ge \sum_t (v_t - 1)$ before the run. Then* STATE-EXP3 *with $m = 1$ and any $\eta \le 1/(e(d+1))$ satisfies $\mathbb{E}[R_T] \le \sqrt{2(3.3V^- + 2dT)\log K} + e(d+1)\log K + d$.*

That is $\widetilde{O}(\sqrt{V^-} + \sqrt{dT})$: the delay is additive, and the algorithm the experiments run is covered, at rates they already satisfy; the proof adapts a deterministic drift induction (Cesa-Bianchi et al., 2019; Thune et al., 2019). Only the first of the three gaps above closes, and the two bounds cross near $\bar{v} = 2$, the additive form gaining above it (Appendix K).

5 The representation trade-off

Some states may not be worth telling apart. Group them with a map g from $\mathcal{S}$ onto a smaller set $\mathcal{Z}$ and run the same algorithm on $g(S_t)$, where $P^g(z \mid a) = \sum_{s:g(s)=z} P(s \mid a)$. Grouping never raises the dimension, but the estimate now tracks a group average, so the widest spread of losses inside a group, $\delta_t(g) = \max_z \max_{s,s':g(s)=g(s')=z} |\theta_t(s) - \theta_t(s')|$, is the error introduced.

Theorem 3. *Under the hypotheses of Theorem 1, running on g with $\sum_t v_t(g) \leq V(g)$ gives $\mathbb{E}[R_T] \leq \sqrt{2(d+1)V(g)\log K} + 2\sum_t \delta_t(g)$.*

Merging states need not preserve state-sufficiency, so the grouped estimate is right not for c_t but for a stand-in loss differing from it by at most $\delta_t(g)$ on every action (Appendix D). Keeping states apart preserves differences that matter but learns more slowly; merging shares more but pays $2\sum_t \delta_t(g)$. The best grouping is the true one: with 16 states from four hidden groups, sizes $1, 2, 4, 8, 16$ give regret 1590, 831, 754, 920, 1080, lowest at four, which no algorithm was told. The map g is fixed in advance; learning it from data remains open.

6 The cost of drift

How much does drift cost even when the stale action losses are free? Theorem 4 shows a cost from time alone, not a joint overlap-drift bound: the construction gives each of the J drifting directions a private state, removing useful pooling and forcing $J \leq |\mathcal{S}| - 1$ (Appendix G).

Theorem 4. *Fix T, a delay $1 \leq d \leq T/2$, a drift budget $\mathcal{E} \leq T/4$, and $1 \leq J \leq \min\{K-1, |\mathcal{S}|-1\}$. Every algorithm, even one handed the exact losses from d rounds ago, meets some instance fixed in advance with $E_2 \leq \mathcal{E}$ on which $\mathbb{E}[R_T] = \Omega(\sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$. Without that help, and if $K \leq T$, the best regret any algorithm can guarantee is $\Omega(\sqrt{KT} + \sqrt{d\,\mathcal{E}\min\{1+\log J,\, T/d\}})$, up to constants.*

The first term is the usual cost of exploring. The second is the cost of delay: it grows with the wait, how far the losses move during it, and the number J of directions the best fixed action can use. They come from different hard instances, so adding them overstates the truth by at most a factor of two.

7 What the experiments show

Pooling gains most when many actions share few state patterns. In a synthetic recommendation funnel with $K = 200$ catalogue items and six engagement states the estimated dimension is 1.97, and the state-pooled variant cuts regret against action-level weighting by 79.3, 63.6 and 38.9 per cent at $d = 10, 50, 200$. It is hand-structured, not fitted to data, so it shows the mechanism, not a deployed system. An online plug-in $\hat{P}_t$ stays within 0.5 per cent of known P (study 1).

K	state m=d+1	state m=1	action-level	paired gain	wins
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

STATE-EXP3 at $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, 100 paired seeds; $\pm$ is one standard error of the paired difference.

These experiments study the $m = 1$ variant, covered by Theorem 2 but not by Theorem 1, whose proof needs $m = d + 1$; Appendix I also evaluates the interleaved variant. A best-state rule leads seven of nine settings, but where its assumption fails its regret grows linearly and its spread across seeds is four times ours. Against a minimax-optimal action-level delayed-bandit baseline the state channel still helps, cutting regret by 32 to 68 per cent on the funnel under the reported tuning protocol (Appendix I).

8 Discussion

The diagnostic is simple: pool when many actions lead to similar states, coarsen when grouped states agree, and distrust delayed information when the state-to-outcome map moves fast.

Conjecture 5 (open). *Some algorithm attains $\mathbb{E}[R_T] = \widetilde{O}(\sqrt{KT} + \sqrt{d\,E_2\log K})$, adapting to both axes.*

References

N. Alon, N. Cesa-Bianchi, O. Dekel, and T. Koren. Online learning with feedback graphs: beyond bandits. *COLT*, PMLR 40, 2015.

P. Auer, N. Cesa-Bianchi, Y. Freund, and R. E. Schapire. The nonstochastic multiarmed bandit problem. *SIAM Journal on Computing*, 32(1):48–77, 2002.

Y. Bar-On and Y. Mansour. Non-stochastic bandits with evolving observations. *ALT*, PMLR 272, 2025.

N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.

N. Cesa-Bianchi, C. Gentile, and Y. Mansour. Delay and cooperation in nonstochastic bandits. *JMLR*, 20, 2019.

K. Eldowa, N. Cesa-Bianchi, A. M. Metelli, and M. Restelli. Information capacity regret bounds for bandits with mediator feedback. *JMLR*, 25(353), 2024.

E. Esposito, S. Masoudian, H. Qiu, D. van der Hoeven, N. Cesa-Bianchi, and Y. Seldin. Delayed bandits: When do intermediate observations help? *ICML*, PMLR 202, 2023.

E. Esposito, F. Fusco, D. van der Hoeven, and N. Cesa-Bianchi. Learning on the edge: online learning with stochastic feedback graphs. *NeurIPS*, 35, 2022.

G. E. Flaspohler, F. Orabona, J. Cohen, S. Mouatadid, M. Oprescu, P. Orenstein, and L. Mackey. Online learning with optimism and delay. *ICML*, PMLR 139, 2021.

G. Guinet, S. Amin, and P. Jaillet. Effective dimension in bandit problems under censorship. *NeurIPS*, 35, 2022.

G. Gabbianelli, G. Neu, and M. Papini. Online learning with off-policy feedback. *ALT*, PMLR 201, 2023.

Y.-G. Hsieh, F. Iutzeler, J. Malick, and P. Mertikopoulos. Multi-agent online optimization with delays. *JMLR*, 23, 2022.

P. Joulani, A. György, and C. Szepesvári. Online learning under delayed feedback. *ICML*, 2013.

M. Mussi, S. Drago, M. Restelli, and A. M. Metelli. Factored-reward bandits with intermediate observations. *ICML*, PMLR 235, 2024.

G. Neu. Explore no more: improved high-probability regret bounds for non-stochastic bandits. *NeurIPS*, 28, 2015.

T. S. Thune, N. Cesa-Bianchi, and Y. Seldin. Nonstochastic multiarmed bandits with unrestricted delays. *NeurIPS*, 32, 2019.

C. Vernade, A. György, and T. Mann. Non-stationary delayed bandits with intermediate observations. *ICML*, PMLR 119, 2020.

S. Wang, A. Barik, and V. Y. F. Tan. Bandit convex optimization with gradient prediction adaptivity. *arXiv:2605.22191*, 2026.

C.-Y. Wei, H. Luo, and A. Agarwal. Taking a hint: How to leverage loss predictors in contextual bandits? *COLT*, PMLR 125, 2020.

K. W. Zhang, T. Baldwin-McDonald, K. Ciosek, L. Maystre, and D. Russo. Impatient bandits: optimizing for the long-term without delay. *JMLR*, to appear, 2026.

M. Zhang. Toward optimal second-order path-length guarantee for adversarial multi-armed bandits. *arXiv:2608.15996*, 2026.

J. Zimmert and Y. Seldin. An optimal algorithm for adversarial bandits with arbitrary delays. *AISTATS*, PMLR 108, 2020.

A Extended related work

Guinet et al. (2022) use the name *effective dimension* for an unrelated quantity, the number of directions a censored-feedback problem can resolve; it is not v_t and the observation model differs.

The minimax characterization of delayed adversarial bandits traces to Cesa-Bianchi et al. (2019), and Joulani et al. (2013) give the additive form of the delay penalty in the stochastic case. On acting before the loss is known, Flaspohler et al. (2021) study optimism under delay directly, Wei et al. (2020) delimit when prediction error controls regret, Hsieh et al. (2022) show that an optimistic step must run at a larger scale than the update, and Zhang (2026) gives second-order path-length bounds.

Structure that lets one observation inform several actions is the other half of the picture. Feedback graphs (Alon et al., 2015) fix which actions are revealed together, and their stochastic version (Esposito et al., 2022) randomizes that set. Factored rewards with intermediate observations (Mussi et al., 2024) decompose the reward instead. Off-policy coverage (Gabbianelli et al., 2023) is the closest analogue of our effective dimension, since its mismatch coefficient measures the same kind of overlap between the distribution that generated the data and the one being evaluated. The distinction from all of these is that a feedback graph randomizes *who* is observed, whereas our sensor randomizes *what* is observed and the sharing is a consequence of P rather than a modeling primitive.

B Notation

Table 1: Symbols used in the paper. Every symbol is also defined at first use in the body, so this table is a reference rather than a dependency.

Symbol	Meaning
T, K, d	rounds, actions, fixed feedback delay
$\mathcal{S}, S_t$	finite state space, state observed at round t
$P \in [0,1]^{K\times\lvert\mathcal{S}\rvert}$	row-stochastic action-to-state matrix, fixed known in Theorems 1 and 3, estimated in Appendix E
$\theta_t \in [0,1]^{\lvert\mathcal{S}\rvert}$	state-loss means at round t, oblivious
$c_t = P\theta_t \in [0,1]^K$	induced loss over actions
$q_t(s) = \sum_a x_t(a)P(s \mid a)$	state distribution induced by the play
$X_t \in [0,1]$	outcome at time $t+d$, $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, A_t, S_t] = \theta_t(S_t)$
x_t, A_t	action distribution at round t, action drawn from it
$m_t = c_{t-d}$	stale action-loss vector, given before the action, $m_t = c_1$ for $t \le d$
E_2	$\sum_t \lVert c_t - m_t\rVert_\infty^2$, accumulated squared prediction error
Λ_2	$\sum_t \min\{1, \lVert c_t - m_t\rVert_2^2\}$, the measure of Bar-On and Mansour (2025)
W	$\sum_t \lVert\theta_t - \theta_{t-1}\rVert_\infty^2$, state-loss variation
$\hat{c}_t, \tilde{c}_t, \check{c}_t$	state-pooled, action-level, centered loss estimates
$\breve{g}_t$	the Safe-Pool blend $(1-\lambda_t)\tilde{c}_t + \lambda_t\hat{c}_t$, a distinct object
$\bar{c}_t, c_t^g, b_t$	conditional mean of $\hat{c}_t$, grouped surrogate loss, $\bar{c}_t - c_t$
$v_t, \bar{v}, V$	effective dimension $\sum_s q_t(s)^{-1}\sum_a x_t(a)P(s \mid a)^2$, its supremum, its budget
$\widehat{\bar{v}}$	Monte Carlo estimate of $\bar{v}$, used for measured values in Appendix I
$g, \mathcal{Z}, \delta_t(g)$	state grouping, its range, widest within-group loss spread
$\mathcal{E}, J$	budget constraining E_2, number of independently drifting coordinates
$h, B, \varepsilon, \sigma_b^{(j)}$	block length, block count, amplitude, Rademacher signs
η, β, C	learning rate, Tsallis parameter, residual scale
α	smoothing pseudocount in $\hat{P}_t$
κ	sensor balance, $\bar{p}(s) \ge 1/(\kappa\lvert\mathcal{S}\rvert)$ for $\bar{p} = K^{-1}\sum_a P(\cdot \mid a)$
m, L_i, G_r	interleaving factor, estimate of copy i, estimates outstanding at r
$D, d_{\max}, \bar{\Lambda}, \Gamma$	total delay, maximal delay, inaccuracy bound, the measure of Wang et al. (2026)

C The state-pooled estimator

The algorithm. Inputs are the action-to-state matrix P, the delay d, the number of copies m, and the rate $\eta > 0$. The analyzed setting is $m = d+1$ and the practical variant is $m = 1$. Round r's outcome becomes usable at round $r+d+1$, which for $m = d+1$ is the next round belonging to the copy that played round r.

1. Set $L_i \leftarrow (0, \dots, 0) \in \mathbb{R}^K$ for $i = 0, \dots, m-1$, and let $\mathcal{P}$ be an empty table of rounds awaiting their outcome.
2. For $t = 1, \dots, T$:
   (a) *Deliver.* If $r \triangleq t - d - 1 \geq 1$, read $(i_r, q_r(S_r), S_r, X_r)$ from $\mathcal{P}$, remove it, and update only that copy, $L_{i_r}(a) \leftarrow L_{i_r}(a) + P(S_r \mid a) X_r / q_r(S_r)$ for every a. This is the pooled step, and it touches every action rather than the one played.
   (b) *Select the copy.* $i \leftarrow t \bmod m$.
   (c) *Play.* $x_t(a) \propto \exp(-\eta L_i(a))$, draw $A_t \sim x_t$, and observe S_t at once.
   (d) *Record.* Compute the scalar $q_t(S_t) = \sum_a x_t(a) P(S_t \mid a)$ and store $(i, q_t(S_t), S_t, \cdot)$ in $\mathcal{P}$, to be completed when X_t arrives at time $t + d$.

Steps (a), (c) and (d) are each $O(K)$ and there is no optimization subproblem, so a round costs $O(K)$ once P is stored. The state is held in the m vectors L_i, which is mK numbers, the matrix P, which is $K|\mathcal{S}|$ numbers, and four scalars for each of the at most d rounds in flight. Only the scalar $q_r(S_r)$ is needed from round r, not the vector x_r.

A worked example. Take three actions and two states, with

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \\ 0 & 1 \end{pmatrix}, \qquad x_t = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}),$$

so $q_t(s_1) = \frac{1}{2}(0.8) + \frac{1}{4}(0.4) + \frac{1}{4}(0) = 0.5$ and $q_t(s_2) = 0.5$. The third action cannot produce s_1 at all. Suppose the learner draws a_1, observes s_1, and receives $X_t = 1$ at time $t + d$. Then $\hat{c}_t(a) = P(s_1 \mid a)/0.5$, so $\hat{c}_t = (1.6, 0.8, 0)$ against the action-level $\tilde{c}_t = (2, 0, 0)$. One observation updates all three actions: a_2 is charged although it was never played, because it had a 0.4 chance of producing the state seen, and a_3 is charged nothing, because it could not have produced that state. Nothing is biased by this. With $\theta_t = (1, 0)$ the true losses are $c_t = (0.8, 0.4, 0)$, and averaging $\hat{c}_t$ over $S_t \sim q_t$ returns exactly that. What is bought is the second moment: $v_t = 1.44$ here, against $K = 3$ for the action-level estimate, and with $\theta_t = (1, 1)$ both are attained exactly.

Remark 6 (The χ^2 form). Under x_t the pair (A_t, S_t) has law $x_t(a) P(s \mid a)$ with marginals x_t and q_t, so

$$\begin{aligned} \chi^2\big(P_{A_t, S_t} \,\big\|\, P_{A_t} \otimes P_{S_t}\big) &= \sum_{a,s} \frac{\big(x_t(a) P(s \mid a) - x_t(a) q_t(s)\big)^2}{x_t(a) q_t(s)} \\ &= \sum_{a,s} \frac{x_t(a) P(s \mid a)^2}{q_t(s)} - 2 + 1 \;=\; v_t - 1, \end{aligned}$$

the cross term summing to $\sum_{a,s} x_t(a) P(s \mid a) = 1$ and the last to $\sum_{a,s} x_t(a) q_t(s) = 1$. So $v_t = 1$ exactly when A_t and S_t are independent, which pins the rows of P only on the support of x_t and therefore means all rows agree whenever x_t has full support, as exponential weights does. If P is deterministic and every state is reached by some action in the support of x_t, then v_t equals the number of states that support reaches. In particular, under full-support play and an onto deterministic P, $v_t = |\mathcal{S}|$. This is an identity, not a bound; it is used for reading v_t, and once in the proof of Theorem 2.

Proposition 7. *Fix t, let x_t be the play distribution, $q_t(s) = \sum_a x_t(a) P(s \mid a)$ the induced state distribution, and $\hat{c}_t(a) = P(S_t \mid a) X_t / q_t(S_t)$, which is well defined because only states with $q_t(s) > 0$ occur. For statements involving $1/x_t(a)$, restrict to actions with $x_t(a) > 0$;* STATE-EXP3 *itself has full support on every action. Then $\hat{c}_t(a) \geq 0$, $\hat{c}_t(a) \leq 1/x_t(a)$, $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$, and*

$$\sum_a x_t(a) \mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\leq\; v_t := \sum_{s: q_t(s) > 0} \frac{\sum_a x_t(a) P(s \mid a)^2}{q_t(s)}, \qquad 1 \leq v_t \leq |\mathcal{S}|,$$

states of zero probability being omitted throughout, which is the convention $0/0 = 0$. The action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\} X_t / x_t(a)$ has the same first three properties and weighted second moment at most K.

Proof. Nonnegativity is immediate from $X_t \geq 0$. Given $\mathcal{F}_{t-1}$ the state S_t has distribution q_t, and $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}, S_t = s] = \theta_t(s)$ by the model, since that conditional mean does not depend on the action, so $\mathbb{E}[\hat{c}_t(a) \mid \mathcal{F}_{t-1}] = \sum_s q_t(s) P(s \mid a)\theta_t(s)/q_t(s) = c_t(a)$. For the range, $q_t(s) \geq x_t(a)P(s \mid a)$ for every a, so $P(s \mid a)/q_t(s) \leq 1/x_t(a)$ and $X_t \leq 1$. For the second moment, $X_t^2 \leq 1$ gives $\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq \sum_s P(s \mid a)^2/q_t(s)$, and exchanging the sums gives v_t. Each summand of v_t is at most 1, since $P(s \mid a) \leq 1$ makes $\sum_a x_t(a)P(s \mid a)^2 \leq q_t(s)$, and at least $q_t(s)$, since Jensen applied to $a \mapsto P(s \mid a)$ under x_t gives $\sum_a x_t(a)P(s \mid a)^2 \geq q_t(s)^2$. Summing the two bounds gives $1 \leq v_t \leq |\mathcal{S}|$. The action-level claims are the same computation with the state replaced by the played action, whose weighted second moment is $\sum_a x_t(a)/x_t(a) = K$. $\square$

The two ends are attained. If every action induces the same state distribution then $v_t = \sum_s q_t(s) = 1$, and if P is deterministic, each action reaching a single state, then every $P(s \mid a)$ is 0 or 1 and each summand is exactly 1, so v_t counts the states reached and equals $|\mathcal{S}|$ when every state has an action reaching it. Disjoint row supports give $v_t = K$ rather than $|\mathcal{S}|$, since the summands over one action's private states add to $\sum_s P(s \mid a) = 1$, so a sensor with many private states per action resolves few of them. So v_t reads the overlap between the rows of P rather than their length, and $\bar{v} = \sup_x v(x)$, the supremum over play distributions, depends on P alone. An outcome observed at a state updates the estimate for every action that can reach it, which is why the construction of Section 6 gives each drifting coordinate a private state, forcing $v_t = |\mathcal{S}|$ there. The immediately observed pairs (A_t, S_t) identify P separately, though Theorem 1 assumes it known.

D Proofs of Theorems 1 and 3

Proof idea. The pooled estimator is unbiased for every action, while Proposition 7 bounds its play-weighted second moment by v_t rather than K. With $m = d+1$ interleaved copies, the outcome generated on a round belonging to one copy is available before that copy acts again, so each copy is an ordinary exponential-weights process with immediate feedback. Summing the $d+1$ potential inequalities gives $(d+1)\log K/\eta$, and the pooled second moments give $\frac{\eta}{2}\sum_t v_t$.

Proof of Theorem 1. Fix a copy i and let T_i be the number of rounds it owns, so $\sum_i T_i = T$. Round t's outcome is usable from round $t+d+1$, which with $m = d+1$ is exactly $t+m$, the copy's next round, so within a copy the feedback is undelayed and L_i carries exactly the estimates of that copy's earlier rounds. Exponential weights on nonnegative losses give, for any action $a^\star$ and any $\eta > 0$,

$$\sum_{t \in i}\big(\langle x_t, \hat{c}_t\rangle - \hat{c}_t(a^\star)\big) \leq \frac{\log K}{\eta} + \frac{\eta}{2}\sum_{t\in i}\sum_a x_t(a)\hat{c}_t(a)^2,$$

by the usual potential telescoping with $e^{-z} \leq 1 - z + z^2/2$, which needs $z \geq 0$ and therefore $\hat{c}_t \geq 0$, supplied by Proposition 7. Each x_t is $\mathcal{F}_{t-1}$-measurable, because L_i then holds only rounds $r \leq t - m = t - d - 1$, so taking expectations turns the left side into $\sum_{t\in i}(\langle x_t, c_t\rangle - c_t(a^\star))$ by unbiasedness and the quadratic term into at most $\mathbb{E}[\sum_{t\in i} v_t]$, again by Proposition 7. Summing over the $m = d{+}1$ copies and using $\sum_i \min_a \sum_{t\in i} c_t(a) \leq \min_a \sum_t c_t(a)$, which is what lets the copies be compared against one common action, gives $\mathbb{E}[R_T] \leq (d+1)\log K/\eta + \frac{\eta}{2}\mathbb{E}[\sum_t v_t]$. If $\sum_t v_t \leq V$ surely then $\eta = \sqrt{2(d+1)\log K/V}$ balances the two terms and gives $\sqrt{2(d+1)V\log K}$, and $V = \bar{v}T$ is admissible by the definition of $\bar{v}$. $\square$

The delay enters as a factor because the copies do not share their estimates, each paying $\log K/\eta$ over $T/(d{+}1)$ rounds. Sharing them is exactly the $m = 1$ algorithm of Conjecture 5, whose analysis is open.

Lemma 8 (Coarsening)**.** *For every play distribution and every g, $v_t(g) \leq v_t$.*

Proof. It suffices to merge two states, the general case following by induction on the merges, and a merged group of zero probability contributes nothing on either side by the convention above. Write P_1, P_2 for the two columns, $A_j = \sum_a x_t(a)P_j(a)^2$, $B_j = \sum_a x_t(a)P_j(a)$ and $\lambda = B_1/(B_1+B_2)$. By Cauchy–Schwarz, $(P_1{+}P_2)^2 \leq \lambda^{-1}P_1^2 {+} (1{-}\lambda)^{-1}P_2^2$ pointwise in a, so $\sum_a x_t(a)(P_1{+}P_2)^2 \leq (B_1+B_2)(A_1/B_1 + A_2/B_2)$, and dividing by the merged denominator B_1+B_2 shows the merged summand is at most $A_1/B_1 + A_2/B_2$, the two it replaces. $\square$

Proof of Theorem 3. Running the algorithm on g is running it on the sensor $(\mathcal{Z}, P^g)$. The surrogate losses below depend on x_t and so are predictable rather than oblivious, but the proof of Theorem 1 uses obliviousness nowhere, only that each c_t^g is $\mathcal{F}_{t-1}$-measurable given the play, so it applies unchanged and bounds regret measured in the surrogate losses $c_t^g(a) = \sum_z P^g(z \mid a)\bar{\theta}_t(z)$, where $\bar{\theta}_t(z) = \sum_{s:g(s)=z} q_t(s)\theta_t(s)/q_t^g(z)$ is the group mean the grouped estimator is unbiased for, by the same computation as in Proposition 7 with S_t replaced by $g(S_t)$. Fix a and z. Both $\{q_t(s)/q_t^g(z)\}$ and $\{P(s \mid a)/P^g(z \mid a)\}$ are probability vectors on $\{s : g(s) = z\}$, so their θ_t-averages lie in a common interval of length $\delta_t(g)$ and differ by at most $\delta_t(g)$. Weighting by $P^g(z \mid a)$ and summing over z gives $|c_t^g(a) - c_t(a)| \le \delta_t(g)$ for every a, hence $\langle x_t, c_t\rangle - c_t(a^\star) \le \langle x_t, c_t^g\rangle - c_t^g(a^\star) + 2\delta_t(g)$. Summing over t adds $2\sum_t \delta_t(g)$, and Lemma 8 makes $V(g)$ no larger than a bound valid for the raw states. □

E An unknown action-to-state map

The plug-in estimator. The pairs (A_t, S_t) arrive without delay, so P is estimable from the learner's own play while the outcomes are still in flight. Let $N_a(t)$ count the rounds before t that played a and $N_{a,s}(t)$ those that also produced s, and set $\hat{P}_t(s \mid a) = (N_{a,s}(t) + \alpha)/(N_a(t) + \alpha|\mathcal{S}|)$ for a smoothing constant $\alpha \in (0, 1]$ fixed independently of T, which is row-stochastic with every entry positive, so $\hat{q}_t = x_t^\top \hat{P}_t$ is positive too, and both are $\mathcal{F}_{t-1}$-measurable. Smoothing moves an entry from the empirical row by at most $\alpha(|\mathcal{S}| - 1)/(N_a(t) + \alpha|\mathcal{S}|)$, attained when every observation of a sits on one state, and moves the row in ℓ_1 by at most twice that. Both are of lower order than the sampling term at fixed α and must be carried otherwise, and Appendix I shows the choice matters in practice. The plug-in estimate is $\hat{c}_t(a) = \hat{P}_t(S_t \mid a)X_t/\hat{q}_t(S_t)$. Write $\varepsilon_t = \max_a \|\hat{P}_t(\cdot \mid a) - P(\cdot \mid a)\|_1$ for its accuracy and $\rho_t = \max_{s:q_t(s)>0} |\hat{q}_t(s) - q_t(s)|/q_t(s)$ for the relative error it induces on the state distribution, states of zero probability under the play being omitted since the estimator never charges them.

Proposition 9. *On $\{\rho_t \le \frac{1}{2}\}$ the plug-in estimate satisfies $\hat{c}_t \ge 0$ and $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \le 2\hat{v}_t \le 2|\mathcal{S}|$, and its conditional mean $\bar{c}_t$ satisfies $\sum_a x_t(a)(\bar{c}_t(a) - c_t(a)) = 0$ exactly, for every $\hat{P}_t$ and with no hypothesis at all, and $|\bar{c}_t(a) - c_t(a)| \le 2\varepsilon_t(a) + 2\rho_t$ for every a, where $\varepsilon_t(a) = \|\hat{P}_t(\cdot \mid a) - P(\cdot \mid a)\|_1$.*

Proof. Write $\Delta_t = \hat{P}_t - P$ and $\delta_t = \hat{q}_t - q_t = \sum_a x_t(a)\Delta_t(\cdot \mid a)$. Substituting these into $\bar{c}_t(a) = \sum_s q_t(s)\hat{P}_t(s \mid a)\theta_t(s)/\hat{q}_t(s)$ and cancelling gives $\bar{c}_t(a) - c_t(a) = \sum_s \theta_t(s)[q_t(s)\Delta_t(s \mid a) - P(s \mid a)\delta_t(s)]/\hat{q}_t(s)$, so with $\theta_t \in [0, 1]^{|\mathcal{S}|}$,

$$|\bar{c}_t(a) - c_t(a)| \;\le\; \sum_s \frac{q_t(s)|\Delta_t(s \mid a)| + P(s \mid a)|\delta_t(s)|}{\hat{q}_t(s)}.$$

The play-weighted identity needs no hypothesis. By definition $\sum_a x_t(a)\hat{P}_t(s \mid a) = \hat{q}_t(s)$, so $\sum_a x_t(a)\hat{c}_t(a) = X_t$ whatever $\hat{P}_t$ is, and taking conditional expectations gives $\sum_a x_t(a)\bar{c}_t(a) = \sum_s q_t(s)\theta_t(s) = \langle x_t, c_t\rangle$. On $\{\rho_t \le \frac{1}{2}\}$ we have $q_t \le 2\hat{q}_t$, and for the per-action bias the first sum of the display is at most $2\|\Delta_t(\cdot \mid a)\|_1 = 2\varepsilon_t(a)$ and the second at most $2\sum_s P(s \mid a)|\delta_t(s)|/q_t(s) \le 2\rho_t$, because $P(\cdot \mid a)$ is a probability vector. The play-weighted identity does *not* survive reweighting: at $x_t = (\frac{1}{2}, \frac{1}{2})$, $P = I$, $\hat{P} = \left(\begin{smallmatrix} .9 & .1 \\ .2 & .8 \end{smallmatrix}\right)$ and $\theta_t = (0, 1)$ the bias is $(\frac{1}{9}, -\frac{1}{9})$ with zero x_t-average, while weights $(1, 0)$ leave $\frac{1}{18}$. For the second moment, $\sum_a x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \le \sum_s \frac{q_t(s)}{\hat{q}_t(s)} \cdot \frac{\sum_a x_t(a)\hat{P}_t(s|a)^2}{\hat{q}_t(s)} \le 2\hat{v}_t$, and $\hat{v}_t \le |\mathcal{S}|$ by Proposition 7 applied to $\hat{P}_t$. □

The warm-start guarantee. WARM-START STATE-EXP3 plays uniformly for n_0 rounds, forms $\hat{P} = \hat{P}_{n_0+1}$ from those rounds alone, discards the cumulative estimates together with every outcome of a warm-start round that has not yet arrived, and then runs STATE-EXP3 with $m = d + 1$, that frozen $\hat{P}$, and each play mixed with the uniform action at rate γ. Freezing makes $\hat{P}$ measurable with respect to the first phase, so ε and ρ are fixed before the second phase begins, and discarding is what keeps the first phase's inaccurate updates out of every later distribution.

Theorem 10. *Suppose $\bar{p}(s) \geq 1/(\kappa|\mathcal{S}|)$ for every s, fix $0 < \gamma \leq \frac{1}{2}$ and $1 \leq d \leq T/2$, and take $n_0 \geq c\,\gamma^{-2} K\kappa^2 |\mathcal{S}|^2 \log(K|\mathcal{S}|T)$ for a universal c. Then* WARM-START STATE-EXP3 *satisfies, for every $\eta > 0$,*

$$\mathbb{E}[R_T] \;\leq\; n_0 + 1 + \gamma T + \frac{(d+1)\log K}{\eta} + 2\eta|\mathcal{S}|T + T\big(4\bar{\varepsilon} + 4\bar{\rho}\big),$$

where $\bar{\varepsilon} = \widetilde{O}(\sqrt{K|\mathcal{S}|/n_0})$ and $\bar{\rho} = \widetilde{O}(\kappa|\mathcal{S}|\sqrt{K/n_0}/\gamma) \leq \frac{1}{2}$ bound ε and ρ on an event of probability at least $1 - 1/T$. Tuning η and then $\gamma \asymp (\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4}$ and $n_0 \asymp T^{4/5}(\kappa|\mathcal{S}|)^{2/5}K^{1/5}$ gives $\mathbb{E}[R_T] = \widetilde{O}\big(\sqrt{d\,|\mathcal{S}|T\log K} + (\kappa|\mathcal{S}|)^{2/5}K^{1/5}T^{4/5}\big)$, provided $T = \widetilde{\Omega}(K(\kappa|\mathcal{S}|)^2)$. That condition constrains T against K, $|\mathcal{S}|$ and κ only, so the hypothesis $d \leq T/2$ is what carries the delay. Together they give $n_0 \leq T/2 \leq T - d$, and the same condition supplies the lower bound above with some $\gamma \leq \frac{1}{2}$. Outside that regime the tuning is vacuous and $R_T \leq T$ is the statement.

Proof. The first phase has losses in $[0,1]$, so it contributes at most n_0. Let G be the event that $N_a(n_0+1) \geq n_0/(2K)$ for every a, that $\|\hat{P}(\cdot \mid a) - P(\cdot \mid a)\|_1 \leq \bar{\varepsilon}$ for every a, and that $\max_{s,a} |\hat{P}(s \mid a) - P(s \mid a)| \leq \bar{\rho}\gamma/(\kappa|\mathcal{S}|)$. Uniform play gives $\mathbb{E}[N_a] = n_0/K$, so a multiplicative Chernoff bound, the ℓ_1 deviation bound for a multinomial, Hoeffding, and a union bound over the $K|\mathcal{S}|$ entries put $\Pr(G^c) \leq 1/T$ at the stated n_0, the pseudo-counts adding $2\alpha(|\mathcal{S}|-1)/(N_a+\alpha|\mathcal{S}|)$ to each ℓ_1 deviation and half that to each entry, of lower order at fixed α. Off G bound the second phase by T, contributing at most 1.

On G the quantity $\hat{P}$ is measurable with respect to the first phase, and mixing gives $q_t(s) \geq \gamma\bar{p}(s) \geq \gamma/(\kappa|\mathcal{S}|)$ at every later round, so $\rho_t \leq \kappa|\mathcal{S}| \max_{s,a} |\hat{P}(s \mid a) - P(s \mid a)|/\gamma \leq \bar{\rho} \leq \frac{1}{2}$ for every $t > n_0$ and every realization of the second phase. Proposition 9 therefore applies at every such round, and conditionally on the first phase the second is exactly the algorithm of Theorem 1 driven by a nonnegative estimate with weighted second moment at most $2\hat{v}_t \leq 2|\mathcal{S}|$. Its proof gives $\mathbb{E}[\sum_{t>n_0}(\langle p_t, \bar{c}_t\rangle - \bar{c}_t(a^\star))] \leq (d+1)\log K/\eta + 2\eta|\mathcal{S}|T$, the extra factor of two coming from $p_t \leq x_t/(1-\gamma) \leq 2x_t$. Playing x_t rather than p_t costs at most γ per round since $c_t \in [0,1]^K$, so it remains to pass from $\bar{c}_t$ to c_t under p_t. Proposition 9 gives $\langle x_t, b_t\rangle = 0$ for the per-action bias b_t, and $x_t = (1-\gamma)p_t + \gamma u$ for the uniform u, so $\langle p_t, b_t\rangle = -\frac{\gamma}{1-\gamma}\langle u, b_t\rangle$, which for $\gamma \leq \frac{1}{2}$ is at most $\max_a |b_t(a)|$ in absolute value. Each round therefore adds $|\langle p_t, b_t\rangle| + |b_t(a^\star)| \leq 2\max_a |b_t(a)| \leq 4\bar{\varepsilon} + 4\bar{\rho}$. Collecting the four contributions gives the display. For the rate, balancing γT against $2T\bar{\rho}$ gives the stated γ and a combined $\widetilde{O}(T(\kappa|\mathcal{S}|)^{1/2}(K/n_0)^{1/4})$, and balancing that against n_0 gives the stated n_0, under which $T\bar{\varepsilon}$ and the constraint on n_0 are of lower order. ☐

Safe blending. Blending the plug-in with the action-level estimate makes the guarantee safe. Let $\lambda_t \in [0,1]^K$ be $\mathcal{F}_{t-1}$-measurable and put $\breve{g}_t(a) = (1-\lambda_t(a))\tilde{c}_t(a) + \lambda_t(a)\hat{c}_t(a)$.

Proposition 11. *$\breve{g}_t \geq 0$, its conditional mean satisfies $\mathbb{E}[\breve{g}_t(a) \mid \mathcal{F}_{t-1}] - c_t(a) = \lambda_t(a)\,(\bar{c}_t(a) - c_t(a))$ for the plug-in's conditional mean $\bar{c}_t$ of Proposition 9, and*

$$\sum_a x_t(a)\mathbb{E}[\breve{g}_t(a)^2 \mid \mathcal{F}_{t-1}] \;\leq\; (K - \Pi_t) + \hat{V}_t,$$

for $\Pi_t = \sum_a \lambda_t(a)$ and $\hat{V}_t = \sum_a \lambda_t(a)x_t(a)\mathbb{E}[\hat{c}_t(a)^2 \mid \mathcal{F}_{t-1}]$, which satisfies $\hat{V}_t \leq 2\hat{v}_t$ on $\{\rho_t \leq \frac{1}{2}\}$ and not in general, since the step $q_t/\hat{q}_t \leq 2$ behind it is exactly that event. We write $V_t^{\mathrm{ub}}(\tau)$ below for a computable upper bound on $\hat{V}_t$, reserving $\hat{V}_t$ for the conditional expectation itself.

Proof. Both parts are nonnegative and $\lambda_t(a) \in [0,1]$. The mean identity is linearity together with $\mathbb{E}[\tilde{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$. For the second moment, $u \mapsto u^2$ is convex, so $\breve{g}_t(a)^2 \leq (1-\lambda_t(a))\tilde{c}_t(a)^2 + \lambda_t(a)\hat{c}_t(a)^2$, and $x_t(a)\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq 1$ by Proposition 7. ☐

So a weight of $\lambda_t(a)$ on the plug-in buys a variance saving of $\lambda_t(a)$ and pays a bias of $\lambda_t(a)$ times the plug-in's, and both endpoints are exact. Let $r_t(a)$ and $\hat{\rho}_t$ be any sequences computable from the immediate pairs that are *simultaneously valid*, meaning $\varepsilon_t(a) \leq r_t(a)$ for every a and $\rho_t \leq \hat{\rho}_t$ at every $t \leq T$, on an event of probability at least $1 - 1/T$. Write $V_t^{\mathrm{ub}}(\tau) = 2\sum_{a:r_t(a)\leq\tau} x_t(a) \sum_s \hat{P}_t(s \mid a)^2/\hat{q}_t(s)$, which the learner can evaluate and which dominates $\hat{V}_t$ on $\{\rho_t \leq \frac{1}{2}\}$. SAFE-POOL

STATE-EXP3 sets $\lambda_t \equiv 0$ when $\hat\rho_t > \frac{1}{2}$, and otherwise picks the best member of the candidate set $\{\perp\} \cup \{r_t(a) : a \le K\}$, where $\perp$ stands for $\lambda_t \equiv 0$ and a numerical candidate τ stands for $\lambda_t(a) = \mathbf{1}\{r_t(a) \le \tau\}$, best meaning smallest

$$\Psi_t(\tau) = \tfrac{\eta}{2}\big[(K - \Pi_t(\tau)) + V_t^{\mathrm{ub}}(\tau)\big] + \big(4\tau + 4\hat\rho_t\big)\mathbf{1}\{\Pi_t(\tau) > 0\}, \qquad \Psi_t(\perp) = \tfrac{\eta}{2}K.$$

Write $\Psi_t^\star$ for that smallest value. The candidate $\perp$ is listed apart from the numerical thresholds because a valid radius may equal zero, in which case $\tau = 0$ pools the actions attaining it rather than abstaining, so $\perp$ rather than $\tau = 0$ is what keeps $\frac{\eta}{2}K$ available on every round.

Theorem 12. *For every $\eta > 0$,* SAFE-POOL STATE-EXP3 *with $m = d + 1$ satisfies*

$$\mathbb{E}[R_T] \;\le\; 1 + \frac{(d+1)\log K}{\eta} + \mathbb{E}\Big[\sum_{t=1}^{T} \Psi_t^\star\Big],$$

so $\eta = \sqrt{2(d+1)\log K/(KT)}$ gives $\mathbb{E}[R_T] \le 1 + \sqrt{2(d+1)KT\log K}$ on every instance, which is the guarantee of action-level exponential weights, with an improvement on each round whose $\Psi_t^\star$ is strictly below $\frac{\eta}{2}K$. The certificate is computed from the learner's own observations and so is random, which is why the display carries an expectation. The tuned bound does not, because $\Psi_t^\star \le \frac{\eta}{2}K$ holds on every path.

Proof. On the validity event, of probability at least $1 - 1/T$ and contributing at most 1 off it, pooling happens only when $\hat\rho_t \le \frac{1}{2}$ and hence only when $\rho_t \le \frac{1}{2}$, which is what makes V_t^{ub} dominate the conditional second moment. Proposition 11 then supplies a nonnegative estimate whose weighted second moment is at most $(K - \Pi_t) + \hat V_t$, so the argument of Appendix D applies to the surrogate losses. Passing to c_t adds both the played bias $|\sum_a x_t(a)\lambda_t(a)b_t(a)|$ and the comparator bias $\lambda_t(a^\star)|b_t(a^\star)|$, for $b_t = \bar c_t - c_t$. The cancellation of Proposition 9 is lost under per-action weights, as the counterexample there shows, so each is bounded separately. On a pooled action $\varepsilon_t(a) \le r_t(a) \le \tau$ gives $|b_t(a)| \le 2\tau + 2\hat\rho_t$, and since $\sum_a x_t(a)\lambda_t(a) \le 1$ and $\lambda_t(a^\star) \le 1$ the two together are at most $4\tau + 4\hat\rho_t$, charged only when something is pooled. The second-moment bound and the two bias terms are all functions of the observations, so the per-round costs they assemble into are the random $\Psi_t^\star$ and the sum is taken in expectation. For the tuned bound, $\perp$ belongs to the candidate set and $\Psi_t(\perp) = \frac{\eta}{2}K$, so $\Psi_t^\star \le \frac{\eta}{2}K$ on every path whatever the radii are. ☐

Why certification stays conservative. The guarantee is safe and the algorithm is inert. Pooling is certified when $4(\tau + \hat\rho_t) \le \frac{\eta}{2}(\Pi_t - V_t^{\mathrm{ub}})$. At the tuned η both sides fall as $T^{-1/2}$, the left through $\tau \asymp \sqrt{K\log(K|\mathcal{S}|T)/T}$ under uniform play and the right through $\eta \asymp \sqrt{\log K/(KT)}$. Their ratio is $\Theta(\sqrt{\log(K|\mathcal{S}|T)/\log K})$, which exceeds one and does not vanish with the horizon, growing slowly in T through the logarithmic factors, so no horizon makes the certificate clear. Appendix I reports behavior consistent with this explanation, the algorithm reproducing action-level regret to the digit with a pooled fraction of zero.

The obstruction is not the shape of the bias bound. Regret charges $\langle x_t, b_t\rangle - b_t(a^\star)$ for the per-action bias b_t, while Proposition 9 charges $\max_a |b_t(a)|$. Measured against the budget itself at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$ and $d = 50$, the first is 0.42 times it and the second 1.67 times it, so on that trajectory the realized bias lies below the best-case variance budget and a tighter bias bound would close the analytical gap. Two exact refinements fail to deliver one. Both $\sum_s \Delta_t(s \mid a)$ and $\sum_s \delta_t(s)$ vanish, so a constant may be subtracted and a maximum replaced by an oscillation, and Cauchy–Schwarz against q_t replaces the comparator term by $\sqrt{\nu_t(a^\star)}\,\chi_t$ for its own effective dimension ν_t and a chi-square χ_t between $\hat q_t$ and q_t. Measured, these are 6.1 and 63 times the truth against 3.9 for the maximum they replace.

What the algorithm can verify from its own counts is 104 times the budget, and Bernstein radii do not improve on Hoeffding ones (Appendix I), so the barrier is certifying ρ_t, the relative error the estimate induces on the state distribution. The play-weighted identity of Proposition 9 says the whole charge is the comparator's, and it routes through ρ_t only because $\hat q_t = x_t^\top \hat P_t$ predicts the state distribution out of every row of P. Predicting is expensive for that reason, needing $t = \Omega(K|\mathcal{S}|\log(K|\mathcal{S}|T)/\rho^2)$ rounds, and the resulting certificate clears for no $K \le T$, since it would need $\log K \ge 16c|\mathcal{S}|$ for $c = \log(2|\mathcal{S}|T)$, which $\log K \le \log T$ cannot meet.

Counting the states costs less and still buys nothing. With x_t frozen over a window the empirical state frequency is a plain multinomial estimate of q_t, needing $n = \Omega(|\mathcal{S}|\log(|\mathcal{S}|T)/\rho^2)$ samples, a factor K fewer than predicting and free of P, and the certificate does clear once $K \log K \geq 64c|\mathcal{S}|$, about $K \geq 10^3$ at $|\mathcal{S}| = 8$. The window that buys it must satisfy $n \geq 32c|\mathcal{S}|T/(K \log K)$, a constant fraction of the horizon, and freezing the play over blocks of length B costs $\sqrt{2TBK \log K}$, which at that B is $T\sqrt{128c|\mathcal{S}|}$ and so linear in T. Both routes fail for one reason: the certificate must resolve q_t to relative accuracy $\Theta(\sqrt{K \log K/T})$, finer than either estimate delivers from the data the learner has.

In words, an unknown action-to-state matrix costs a bias in two accuracies, the ℓ_1 error of $\hat{P}$ and the relative error it induces on the state distribution, and only the second needs forced exploration, which is what sets the rate. **The bound does not certify an improvement from the channel it analyzes.** Its second term exceeds $\sqrt{KT}$ at every horizon with $T > K$, by a factor $(T/K)^{3/10}$, so a learner holding this guarantee alone should run action-level exponential weights instead. The experiments suggest the analysis is conservative rather than the algorithm weak, since Appendix I shows it tracking known P closely, and sharpening it is the natural next result.

F Feedback models, and what transfers

Write $\Lambda_2 = \sum_t \min\{1, \|c_t - m_t\|_2^2\}$ for the inaccuracy measure of Bar-On and Mansour (2025). Unlike E_2, which charges only the worst action in a round, Λ_2 grows with how many actions are predicted wrongly, and since $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{J}\|v\|_\infty$ for a vector v with J nonzero coordinates, $E_2 \leq \Lambda_2 \leq J\,E_2$ with both ends attained.

Bar-On and Mansour (2025) analyze a model in which the played coordinate of the loss vector is observed exactly. Section 3 instead observes a random outcome whose conditional mean is the loss. The following records precisely what survives the substitution.

Proposition 13. *Take the action-level $\tilde{c}_t(a) = \mathbf{1}\{A_t = a\}X_t/x_t(a)$ of Proposition 7 under the model of Section 3. Then for every action a, (i) $\mathbb{E}[\tilde{c}_t(a) \mid \mathcal{F}_{t-1}] = c_t(a)$, (ii) $0 \leq \tilde{c}_t(a) \leq 1/x_t(a)$, and (iii) $\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] \leq 1/x_t(a)$. The centered estimator $\check{c}_t(a) = m_t(a) + \mathbf{1}\{A_t = a\}(X_t - m_t(a))/x_t(a)$ also satisfies (i), and its residual obeys $\mathbb{E}[(\check{c}_t(a) - m_t(a))^2 \mid \mathcal{F}_{t-1}] \leq 1/x_t(a)$.*

Proof. For (i), by the tower rule $\mathbb{E}[\mathbf{1}\{A_t = a\}X_t \mid \mathcal{F}_{t-1}] = x_t(a)\,\mathbb{E}[X_t \mid A_t = a] = x_t(a)c_t(a)$, and dividing by $x_t(a)$ gives the claim. Item (ii) holds because $X_t \in [0,1]$. For (iii), $\mathbb{E}[\tilde{c}_t(a)^2 \mid \mathcal{F}_{t-1}] = \mathbb{E}[X_t^2 \mathbf{1}\{A_t = a\}]/x_t(a)^2 = \mathbb{E}[X_t^2 \mid A_t = a]/x_t(a) \leq 1/x_t(a)$ because $X_t \leq 1$. The statements for $\check{c}_t$ follow the same computation with X_t replaced by $X_t - m_t(a)$, using $|X_t - m_t(a)| \leq 1$. $\square$

In words, replacing an exactly observed loss by a bounded unbiased draw of it changes neither the mean of the estimator nor the moment bounds an analysis consumes.

Under exact feedback the corresponding estimator satisfies (i) to (iii) with identical bounds, since $c_t(a) \leq 1$. Consequently any step of an exact-feedback analysis that uses the observation only through unbiasedness and these two moment bounds is unchanged under noisy outcomes. What this does not cover is the comparison between the estimate built from the stale action-loss vector and the estimate built from the true loss. Under exact feedback that difference is determined by the played action. Under noisy outcomes it carries an additional mean-zero term of magnitude up to $1/x_t(A_t)$, and bounding its cumulative effect over a window of d rounds is what requires the stability property falsified in Appendix H. An upper bound for the noisy model is Theorem 1. What remains open is one adaptive to E_2.

G Proofs

The measure E_2 is controlled by the state-loss variation W, and no converse bound holds in general.

Lemma 14 (Delay window)**.** *$E_2 \leq d^2 W$ for $W = \sum_t \|\theta_t - \theta_{t-1}\|_\infty^2$. The ratio $E_2/(d^2 W)$ tends to one when θ drifts monotonically in one coordinate at a constant rate and $T/d \to \infty$, the first d rounds contributing $\sum_{j<d} j^2$ rather than d^2 each under the convention $m_t = c_1$.*

In words, accumulated squared prediction error is at most the state-loss variation inflated by the square of the delay, and no better bound in W alone is available.

Proof of Lemma 14. Throughout write $\Delta_j = \theta_j - \theta_{j-1}$, set $\theta_0 = \theta_1$, so $\Delta_1 = 0$, and read c_r as c_1 for $r \le 0$, matching the convention $m_t = c_1$ for $t \le d$ of Section 3. Then $c_t - c_{t-d} = P \sum_{j=t-d+1}^{t} \Delta_j$. Because the rows of P are probability vectors, averaging cannot increase the supremum norm, so $\|c_t - c_{t-d}\|_\infty \le \sum_j \|\Delta_j\|_\infty$. By Cauchy–Schwarz applied to the d summands, $\|c_t - c_{t-d}\|_\infty^2 \le d \sum_{j=t-d+1}^{t} \|\Delta_j\|_\infty^2$. Summing over t and using that each increment belongs to at most d windows gives $E_2 \le d^2 W$. The ratio $E_2/(d^2 W)$ approaches one when every increment has the same magnitude, sign and active coordinate, and some action reaches that coordinate deterministically, since then no cancellation occurs within a full window and P passes the increment through undiminished. Without the second condition the drift can lie in the kernel of P and the ratio can be zero. □

Lemma 15 (Information isolation)**.** *If $h \le d$ then A_t is independent of the signs governing its own block, and this remains true when the learner is provided with the exact stale action-loss vector m_t.*

Proof of Lemma 15. For t in block b, every realized outcome usable when choosing A_t comes from a round $r \le t-d-1$. Since $t \le bh$ and $h \le d$, we have $r \le bh - d - 1 < (b-1)h$, so every usable realized outcome depends only on signs from blocks strictly before b, whose joint law depends only on $\sigma_2, \dots, \sigma_{b-1}$. Hence by conditional independence A_t is independent of block b's signs. For the oracle, take any $h \le d$. If $t > d$ then $t - d \le bh - d \le (b-1)h$ by the same computation, so $m_t = c_{t-d}$ is measurable with respect to the signs of blocks strictly before b. If $t \le d$ then $m_t = c_1 = \frac{1}{2}\mathbf{1}$, which is deterministic because block 1 is neutral. In both cases m_t is independent of block b's signs, so adjoining it to the learner's information leaves the conclusion intact. □

The construction in full. Take states $s_0, \dots, s_q$, a stationary deterministic map sending a_j to s_j for $j \le J$ and every other action to s_0, and independent Rademacher signs $\sigma_b^{(j)}$ for $2 \le b \le B = \lfloor T/h \rfloor$. Set $\theta_t(s_0) = \frac{1}{2}$ throughout, $\theta_t \equiv \frac{1}{2}$ on block 1, and $\theta_t(s_j) = \frac{1}{2} - \varepsilon\sigma_b^{(j)}$ on each later block b. The neutral first block makes $c_1 = \frac{1}{2}\mathbf{1}$, so the convention $m_t = c_1$ for $t \le d$ of Section 3 supplies a constant over exactly the rounds that precede any usable feedback, which is what Lemma 15 needs at $b = 1$. Rounds after the last complete block carry no drift, costing a constant factor.

Lemma 16 (Rademacher lower tail)**.** *Let S_B be a sum of $B \ge 32$ independent Rademacher signs. For every $1 \le u \le \sqrt{B}/32$, $\Pr(S_B \ge u\sqrt{B}) \ge \frac{u}{4}e^{-64u^2}$.*

Proof. Write $S_B = 2X - B$ with $X \sim \mathrm{Bin}(B, \frac{1}{2})$, so that $\{S_B \ge u\sqrt{B}\} = \{X \ge B/2 + u\sqrt{B}/2\}$. Passing to X removes the lattice gap, since X takes every integer value in $[0, B]$ whereas S_B takes only those of the parity of B. Put $n = \lfloor B/2 \rfloor$ and $t = \lceil u\sqrt{B}/2 \rceil + 1$, so $X \ge n + t$ implies the event, and $t \le u\sqrt{B}$ because $u\sqrt{B} \ge \sqrt{32} > 4$.

For $j \ge 1$ the ratio of consecutive binomial probabilities is $\Pr(X = n+j)/\Pr(X = n+j-1) = (B-n-j+1)/(n+j)$. Using $(B-1)/2 \le n \le B/2$, its distance from one is at most $(2j-1)/(n+j) \le (4j-2)/(B-1) \le 8j/B$ for $B \ge 2$. Restricting to $j \le B/16$ keeps $8j/B \le \frac{1}{2}$, where $\log(1-x) \ge -2x$ holds, so summing over $i \le j$ gives $\Pr(X = n+j) \ge \Pr(X = n)e^{-16j(j+1)/(2B)} \ge \Pr(X = n)e^{-16j^2/B}$. The central term satisfies $\Pr(X = n) \ge 1/(2\sqrt{B})$ for every $B \ge 2$.

Sum over the t integers $j \in \{t, \dots, 2t-1\}$, all of which are at most $2t$ and hence at most $B/16$ because $u \le \sqrt{B}/32$ forces $2t \le 2u\sqrt{B} \le B/16$. Each term is at least $e^{-64t^2/B}/(2\sqrt{B})$, and $t \le u\sqrt{B}$ makes the exponent at most $64u^2$, so $\Pr(X \ge n+t) \ge t\,e^{-64u^2}/(2\sqrt{B}) \ge \frac{u}{4}e^{-64u^2}$ using $t \ge u\sqrt{B}/2$. □

Three regimes give the maximal inequality the lower bound needs. Each $S_B^{(j)}$ is sub-Gaussian with variance proxy B, so $\mathbb{E}\max_j (S_B^{(j)})^+ \le \sqrt{2B\log J} + \sqrt{B}$ throughout. For the matching lower bound, first suppose $\log J \le B/8$ and $\log J \ge 128$. Taking $u = \sqrt{\log J/128}$, which then lies in $[1, \sqrt{B}/32]$, makes $e^{-64u^2} = J^{-1/2}$, so $\Pr(S_B^{(j)} \geq u\sqrt{B}) \geq \frac{u}{4} J^{-1/2}$ and independence across j gives $\Pr(\max_j S_B^{(j)} < u\sqrt{B}) \leq e^{-u\sqrt{J}/4}$, at most $\frac{1}{2}$ beyond an absolute J. Hence $\mathbb{E} \max_j (S_B^{(j)})^+ \geq \frac{1}{2} u\sqrt{B} = \Omega(\sqrt{B \log J})$. Second, if $\log J > B/8$ and $B \geq 1024$, take $u = \sqrt{B}/32$, so $u\sqrt{B} = B/32$ and $e^{-64u^2} = e^{-B/16}$, and $J \geq e^{B/8}$ makes $J\, e^{-B/16}$ diverge, giving $\mathbb{E} \max_j (S_B^{(j)})^+ = \Omega(B)$, which is $\Omega(\sqrt{B \min\{1 + \log J, B\}})$ in that regime. Third, when $\log J < 128$ or $B < 1024$ the quantity $\sqrt{B \min\{1 + \log J, B\}}$ is $O(\sqrt{B})$, and a single walk already gives $\mathbb{E}(S_B^{(1)})^+ = \frac{1}{2}\mathbb{E}|S_B| \geq \frac{1}{2}(\mathbb{E}S_B^2)^{3/2}(\mathbb{E}S_B^4)^{-1/2} \geq \frac{1}{2}\sqrt{B/3}$ by Hölder, since $\mathbb{E}S_B^2 = B$ and $\mathbb{E}S_B^4 \leq 3B^2$. Combining the three regimes, $\mathbb{E} \max_j (S_B^{(j)})^+ = \Theta(\sqrt{B \min\{1 + \log J, B\}})$. This is the route by which maxima of independent random walks fix the minimax rate for prediction with expert advice (Cesa-Bianchi and Lugosi, 2006).

Proof idea. Divide time into blocks of length d and give J actions private states whose losses are perturbed by independent Rademacher signs within each block. Every observation available when an action is chosen comes from an earlier block, and the supplied $m_t = c_{t-d}$ depends only on earlier signs, so the learner cannot predict the signs governing its current block. Its expected loss is therefore $T/2$, while the best fixed action in hindsight benefits from the maximum of J independent random walks. Choosing the amplitude to make $E_2 \leq \mathcal{E}$ gives the stated scale.

Proof of Theorem 4. Take $h = d$ and $\varepsilon = \frac{1}{2}\sqrt{\mathcal{E}/T}$. By Lemma 15 every algorithm's expected loss equals $T/2$, since its action is independent of the current block's signs and all states have mean $\frac{1}{2}$ marginally. Its regret is therefore exactly the comparator's fluctuation advantage, $\varepsilon d\, \mathbb{E}\big[\max_{j \leq J} \big(\sum_{2 \leq b \leq B} \sigma_b^{(j)}\big)^+\big]$ with $B = \lfloor T/d \rfloor$, over the $B - 1$ randomized blocks. Per round $\|c_t - m_t\|_\infty \leq 2\varepsilon$, because the supremum norm charges only the largest coordinate and each coordinate moves by at most 2ε at a block boundary; the gap is ε across the first randomized block, where m_t is still the neutral c_1, and lies in $\{0, 2\varepsilon\}$ thereafter. Hence $E_2 \leq 4\varepsilon^2 T = \mathcal{E}$ holds deterministically. The budget must bind on the realized sequence, since a budget holding only in expectation would not constrain the instance selected by Yao's principle. By Lemma 16 and the discussion following it, the expected maximum of the positive parts of J independent Rademacher sums of length $B-1$ is $\Theta(\sqrt{(B-1)\min\{1 + \log J, B-1\}})$, which covers both regimes and the case $J = 1$. The hypothesis $d \leq T/2$ gives $B \geq 2$ and hence $B - 1 \geq B/2$, so replacing $B - 1$ by B costs a factor at most $\sqrt{2}$ in each branch of the minimum, including the saturated branch where the minimum is $B-1$ rather than $1 + \log J$. Substituting $B = \lfloor T/d \rfloor$ and ε therefore gives $\Theta(\sqrt{d\mathcal{E} \min\{1 + \log J, T/d\}})$. Rounds after the last complete block carry no drift and change the constant only. When the oracle is withheld, the $\sqrt{KT}$ term is Esposito et al. (2023, Thm. 5.1), whose construction is stationary and therefore lies in the present model class. ☐

Proposition 17. *Suppose $X_t = c_t(A_t)$. On instances whose drift is carried by a single action, $\Lambda_2 = E_2$, so the bound of Bar-On and Mansour (2025) reads $\widetilde{O}(\sqrt{KT + d\, E_2})$ and its drift contribution $\sqrt{d\, E_2}$ matches Theorem 4 up to logarithmic factors.*

In words, single-action drift makes the Euclidean and supremum measures agree, so on that family the drift term of the published upper bound is unimprovable in order. Its $\sqrt{KT}$ term is not matched because the oracle supplies an entire stale loss vector, so the oracle-assisted problem need not retain bandit exploration complexity. Neither result provides an E_2-adaptive upper bound under noisy outcomes.

Proof of Proposition 17. If only one coordinate of $c_t - m_t$ is nonzero then $\|c_t - m_t\|_2 = \|c_t - m_t\|_\infty$. On the family of Section 6 that coordinate has magnitude at most $2\varepsilon \leq 1$, so the truncation at one in Λ_2 is inactive and $\Lambda_2 = E_2$. The bound of Bar-On and Mansour (2025) then reads $\widetilde{O}(\sqrt{KT + dE_2})$, whose $\sqrt{dE_2}$ contribution Theorem 4 matches up to the logarithmic factor at $J = 1$. Its $\sqrt{KT}$ contribution is not matched. ☐

H The two-action counterexample

Take $K = 2$ with the hybrid regularizer of Zimmert and Seldin (2020), Tsallis parameter $\beta = 0.01$, learning rate $\eta = 0.5$ and residual scale $C = 1$. The iterate solves the stationarity condition $g(x_1) + L_1 = g(x_2) + L_2$ with $x_1 + x_2 = 1$, where

$$g(x) = -\tfrac{1}{2\beta}x^{-1/2} + \tfrac{1}{\eta}\log x$$

and L_a is the cumulative estimate for action a. Suppose the cumulative estimates place action 1 at probability x_1. A single maximally negative centered residual $-C/x_1$ is then delivered to action 1. This is realizable in the model of Section 3, since $m_t(1) = 1$ and $X_t = 0$ give exactly $\check{c}_t(1) - m_t(1) = -C/x_1$ with $C = 1$. Solving the stationarity condition again gives the following.

x_1 before	x_1 after	ratio
9.15×10^{-3}	1.46×10^{-2}	1.6
9.83×10^{-4}	7.39×10^{-3}	7.5
2.19×10^{-4}	9.98×10^{-1}	4561
5.80×10^{-5}	1.00	17226
4.86×10^{-6}	1.00	205641

The transition sits at $4\beta^2C^2 = 4 \times 10^{-4}$, which is where the Tsallis term $x^{-1/2}$ ceases to dominate the entropy term $\log x$ in g, so the one-step linearization underlying a constant-factor bound fails. The violated statement is the local-stability inequality asserting that $x_{t'}(a)/x_t(a)$ stays bounded by a constant for all t' within d rounds of t. A single round refutes it, so no accumulation argument is needed. This invalidates that lemma and the proof route resting on it, and not every argument that might use the same regularizer. It does not refute Conjecture 5, which concerns the regret rate and may be provable by other means. Devices that bound the residual magnitude introduce either a first-order bias dominating the target term or a probability floor narrowing the delay regime.

I Numerical detail

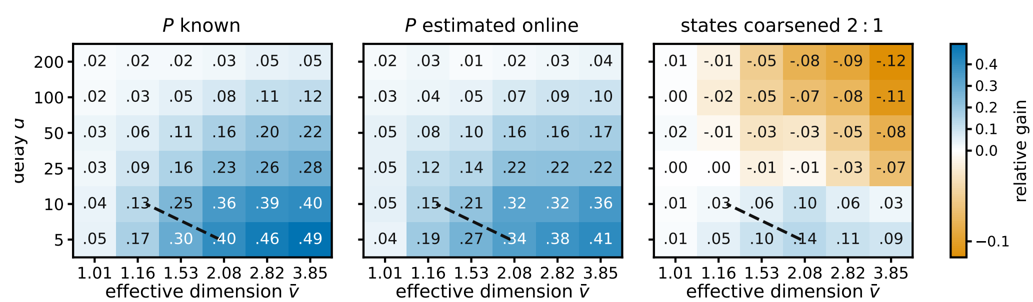

Figure 2: Relative gain $(R_{\text{action}} - R_{\text{state}})/R_{\text{action}}$ of pooling through the state, for the practical $m = 1$ variant, at $K = 40$, $|\mathcal{S}| = 8$, $T = 5000$ and 12 paired seeds. Blue is a gain, and $\widehat{\tilde{v}}$ denotes a Monte Carlo estimate of $\widehat{\tilde{v}} = \sup_x v(x)$ obtained from 4000 sampled play distributions of the *raw* map, which for the coarsened panel is not the $\bar{v}(P^g)$ that Theorem 3 charges. The dashed line marks where the bound of Theorem 1, which analyzes $m = d + 1$, falls below $\widetilde{O}(\sqrt{(K+d)T})$. It compares two upper bounds for a variant other than the one plotted, so it is a reference and not a prediction.

Sixteen studies, alongside Figure 2, which sweeps row overlap against delay in the three settings the studies treat separately, P known, P estimated, and states coarsened. Studies one to thirteen draw P from a Dirichlet and studies fourteen to sixteen do not throughout. STATE-EXP3 is beaten in study fifteen, and in four of six cells of study sixteen when the baseline alone is grid-tuned. Each study reports an observation, an interpretation, and a limitation. Studies 8 and 9 ask whether the two theorems describe the right *dependence* rather than whether a bound holds, which is the question a loose bound leaves open. Study 10 is a control for study 6 and reverses its conclusion, and study 13 reports a rule that does not work. All policies run on the same environment parameters and common random-number streams, so a seed fixes everything except which estimator is formed, and differences are taken within a seed before averaging. Round r's outcome is consumed before round $r + d + 1$, matching Section 3. Reported $\pm$ is one standard error of the paired difference. Throughout this appendix, *sample-path pseudo-regret* means $\sum_t c_t(A_t) - \min_a \sum_t c_t(a)$, evaluated on one realization of the learner and environment randomness. Averaging it over seeds estimates the expected pseudo-regret that the theorems bound. We use that name rather than "realized regret" to distinguish it from regret computed from the sampled outcomes X_t, which is not what these studies report.

Does the bound hold, and does the estimator help? With $|\mathcal{S}| = 4$, $T = 10000$, $d = 50$, drift band 0.05 and 100 seeds:

K	state, $m = d+1$	state, $m = 1$	action, $m = 1$	paired gain	state better in
8	455.1	368.2	387.6	19.3 ± 3.4	73/100
20	618.3	493.7	547.5	53.8 ± 5.6	87/100
40	734.3	574.5	696.1	121.5 ± 9.4	99/100
80	780.4	614.8	784.0	169.3 ± 9.9	100/100

Observed. Sample-path pseudo-regret for the analyzed variant stayed below $\sqrt{2(d+1)|\mathcal{S}|T \log K}$ in 100 of 100 seeds at every K, the bound running from 2913 to 4228 against realized 455 to 780. The plain variant beat the action-level baseline at every K, by a margin growing from 19 to 169 as K rose tenfold with $|\mathcal{S}|$ fixed, and it beat the analyzed variant throughout. Sweeping the delay at $K = 40$ gave state-pooled 387, 569 and 731 against action-level 653, 699 and 773 at $d = 10$, 50 and 200.

Interpretation. The margin growing in K at fixed $|\mathcal{S}|$ is the prediction that separates the two estimators, so the improvement is attributable to the state channel the effective dimension attributes to it. Interleaving looks like a price of the analysis rather than of the estimator. The narrowing of the gap as the delay grows is what an additive delay term predicts and a multiplicative one does not, which is the diagnostic evidence behind Conjecture 5.

Limitation. This is one synthetic family with a fixed drift band, the plain variant carries no guarantee, and a simulation cannot establish that a bound holds, only that sample-path pseudo-regret stayed below it in the runs performed.

What does an unknown map cost? What follows evaluates a practical variant, the plug-in estimate with $\hat{P}_t$ updated every round, no restart, $m = 1$ and $\alpha = 0.01$, and not the warm-start algorithm of Theorem 10, which freezes $\hat{P}$ and restarts. Against the same algorithm given P and against the action-level baseline, over 60 paired seeds and at mixing rate γ:

K	γ	known P	plug-in	action-level	plug-in gain
8	0.00	366.2	365.3	385.4	20.0 ± 4.3
20	0.00	480.3	477.9	529.0	51.2 ± 7.6
40	0.00	582.8	585.3	710.2	124.8 ± 11.2
80	0.00	625.2	623.1	802.2	179.1 ± 12.9
40	0.02	590.5	592.8	712.9	120.1 ± 10.4
40	0.10	618.9	623.5	735.5	112.0 ± 10.1

Observed. The plug-in landed within 0.5 per cent of known P at every K and beat action-level weighting by 20 to 179 as K rose tenfold. Forced exploration did not help, $\gamma = 0$ being best at every K. At $K = 40$ the sampled maps have $\kappa \approx 1.14$, so the exploration term $(\kappa|\mathcal{S}|)^{2/5} K^{1/5} T^{4/5}$ of Theorem 10 evaluates to about 6100 and its leading term to about 2700, against sample-path pseudo-regret 585 and against $\sqrt{KT} = 632$.

Interpretation. Estimating P is nearly free on this family, and the gap between the analysis and the behavior is an order of magnitude, so the analysis rather than the method looks conservative.

Limitation. The variant run here has no guarantee of its own, so the comparison is against a rate proved for a different algorithm. That $\gamma = 0$ wins is a statement about this family rather than about the algorithm, which the next study tests.

Where does the plug-in fail? The family above is mild. Five deliberately hostile maps at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 30 paired seeds, with κ measured on each: a best action reachable only through a state the others rarely produce, a sensor whose mass sits on one state, a state of probability 10^{-3} everywhere, a deterministic map, and one whose good action's early visits are unrepresentative.

				plug-in at mixing rate γ			
environment	κ	known P	action-level	0	0.01	0.05	0.20
rare optimal action	125.9	950.7	1302.3	948.2	976.8	1087.8	1503.0
unbalanced coverage	152.7	483.4	604.4	474.1	476.2	487.0	531.8
near-unreachable state	108.8	552.9	650.4	559.5	563.3	575.8	624.7
deterministic rows	2.5	556.3	733.7	571.9	585.9	653.9	904.3
misleading early visits	1.5	674.0	847.2	679.9	688.0	716.8	834.3

Observed. The plug-in beat action-level weighting on all five, including two with κ above 100, and forced exploration cost regret in every cell. At the Laplace default $\alpha = 1$ the deterministic map is the one case where the plug-in *loses* to action-level weighting, 750.3 against 733.7. Sweeping α over 1, 0.3, 0.1, 0.03, 0.01 gives 750.3, 642.5, 601.4, 581.1, 571.9 there and is monotone on all five, and $\alpha = 0.01$ is what the tables above use.

Interpretation. The mixing rate that Theorem 10 tunes is doing analytic work rather than algorithmic work. The single loss has an identifiable mechanism, since one visit to a deterministic row is recorded as $\hat{P}(s \mid a) = 2/(1 + |\mathcal{S}|)$ rather than 1, so rarely played actions are scored as if spread over states they never reach.

Limitation. The leading rate is indifferent to fixed α and the algorithm is not, which is worth stating rather than tuning away. These five maps show that hostility to the plug-in is survivable at a small constant, not that it is harmless, and no map adversarial to the smoothing constant itself was constructed.

Does safe blending recover the pooling gain, and if not why not? The rule of Theorem 12, run in the $m = 1$ variant like everything else in this appendix rather than in the analyzed $m = d + 1$ one, was compared against its two ingredients at $K = 40$, $|\mathcal{S}| = 8$, $d = 50$ and 20 paired seeds. Measuring the realized bias against the certificate's own budget locates the outcome. At $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 12 seeds the per-round budget $\frac{\eta}{2}(K - 2|\mathcal{S}|)$ sums to 307.3 over the horizon, and

summed	versus budget	quantity
128.7	0.42×	what the regret charges, $\sum_t |\langle x_t, b_t\rangle - b_t(a^\star)|$
513.7	1.67×	what Proposition 9 charges, $\sum_t \max_a |b_t(a)|$
32000.0	104×	what the algorithm can verify, Hoeffding radii
32000.0	104×	the same with Bernstein radii

Observed. The rule reproduced action-level regret to the digit in every cell, 533.2, 754.4 and 1317.0 at $T = 8000$ on the Dirichlet, deterministic and rare-optimal families and 2304.6, 2023.5 and 3108.8 at $T = 50000$, against a plug-in that scores 464.8, 585.5, 956.5 and 1770.6, 1473.5, 2346.6 on the same runs. The fraction of round-action pairs it cleared for pooling was 0.000 everywhere, at both horizons, at both smoothing constants, and with the confidence logs dropped as well.

Interpretation. The implemented certificate selected no pooling in any run, so the policy coincided numerically with the action-level baseline rather than merely matching its guarantee, and the table says where that comes from. On this trajectory the realized bias lies below the best-case variance budget, so pooling would pay for itself here. The bound of Proposition 9 is 1.67 times the budget with a factor of 4.0 available between it and the truth, so a tight bias bound would close the analytical gap. What the algorithm can verify is saturated at the trivial cap of four per round, and Bernstein radii buy nothing over Hoeffding ones, which is where the failure lives.

Limitation. Two exact refinements of the bias bound, an oscillation in place of a maximum and a Cauchy–Schwarz split into the comparator's effective dimension times a chi-square, give 577.9 and 5931.6 against a truth of 94.1 over $T = 4000$, where the maximum gives 368.0, so both are worse than what they replace. The budget table is one diagnostic trajectory rather than a worst case, so it shows this certificate inert here and not that no certificate can clear.

Can the denominator be counted instead of predicted? With x frozen over blocks of 400 and the states of the first half counted, at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 12 seeds, the counted denominator gives regret 496.5 against 499.5 for the predicted one and a comparator bias of 99.6 against 52.9, both under the budget of 153.7 for the pooled half.

Observed. Counting is the noisier of the two on regret and on bias. Computed under uniform play, the predicted relative error is vacuous at every setting tried, $K = 40$ through $K = 4000$, while the counted one reads 0.0244 against a budget of 0.0892 at $K = 1000$ and $T = 10^5$.

Interpretation. A prediction pools every action's history while a count uses one block, which is why counting is noisier, and yet the counted certificate is the better scaled of the two. That setting is the first in which any certificate for ρ_t clears at all.

Limitation. The window that buys it is 53 per cent of the horizon, and the frozen-block regret at that window, 1.2×10^7, exceeds the 2.6×10^4 the pooling was meant to improve, so what clears is the certificate and not a usable algorithm.

Does the effective dimension, rather than $|\mathcal{S}|$, govern the gain? Holding $|\mathcal{S}| = 8$, $K = 40$, $T = 5000$, $d = 20$ and 24 seeds, and varying only the overlap between the rows of P, the measured mean of v_t moves while $|\mathcal{S}|$ does not:

row overlap	mean v_t	state-pooled	action-level	paired gain
0.00	2.719	392.7	599.3	206.6 ± 20.0
0.25	2.013	361.1	528.1	167.0 ± 21.3
0.50	1.532	306.5	406.5	100.0 ± 10.5
0.75	1.168	199.9	229.2	29.3 ± 3.2
0.95	1.009	49.2	50.8	1.6 ± 0.5

Observed. Both the regret and the advantage over action-level weighting fall monotonically with v_t, from a paired gain of 206.6 at $v_t = 2.719$ to 1.6 at $v_t = 1.009$. Nothing about the raw state count changes across the sweep.

Interpretation. $|\mathcal{S}|$ is constant and cannot explain the pattern, while v_t moves with it, which is the comparison Theorem 1 predicts.

Limitation. The two estimators do not become the same object at $v_t \approx 1$. Rather the rows of P agree, so every action induces the same loss and both regrets collapse, making that endpoint uninformative rather than favorable. One parameter of one map family was varied, so the sweep is consistent with the effective dimension governing the gain rather than establishing it.

Is there a bias-variance trade in the representation? With 16 raw states drawn from four latent clusters, $K = 40$, $T = 6000$, $d = 20$ and 24 seeds, running the algorithm on nested groupings of size $1, 2, 4, 8, 16$ gives, for the analyzed $m = d + 1$ variant, regret 1590, 831, 754, 920, 1080 at within-cluster heterogeneity 0, with mean v_t rising 1.000, 1.122, 1.435, 1.663, 2.089 and $\delta_t(g)$ falling 0.700, 0.275, 0, 0, 0.

Observed. The minimum sits at four groups, the number of latent clusters, and stays there at heterogeneity 0.12, where regret reads 1657, 885, 787, 960, 1113. Sample-path pseudo-regret stayed below the bound of Theorem 3 in every cell. The bound's own optimum agrees at heterogeneity 0, where it reads 9364, 4321, 1155, 1243, 1394 and also selects four groups, but not at 0.12, where it reads 11128, 6073, 3430, 3231, 1385 and selects the finest representation.

Interpretation. A trade-off in the representation exists and is located at the latent structure, which is what Theorem 3 describes qualitatively.

Limitation. The bias term $2\sum_t \delta_t(g)$ is the conservative half of the trade, so the theorem certifies a trade-off whose location it does not predict once groups are heterogeneous, and the latent cluster count was known to the experimenter and not to any algorithm.

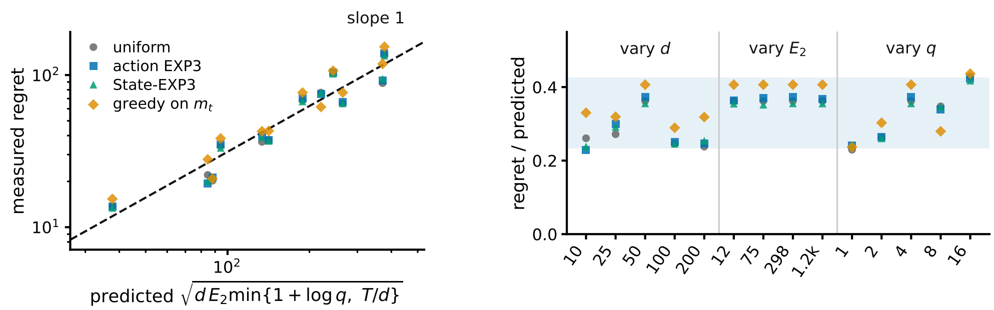

Figure 3: Study 8, the drift channel. **Left,** measured regret against the scale Theorem 4 predicts, over twelve cells in which d varies 20-fold, E_2 100-fold and J 16-fold. The dashed line has slope one. **Right,** the same points divided by that scale, grouped by which parameter the cell sweeps, with the shaded band spanning the observed range. STATE-EXP3 appears here only as one more algorithm on the hard family, where it has no advantage.

Separately, on the construction of Theorem 4 with $h = d$ the mean pseudo-regret over 400 seeds is 44.1 against a predicted fluctuation of 40.5, and of the five policies compared there the four without access to the block signs attain expected loss within 0.05% of $T/2$, as Lemma 15 predicts, while the one given them attains pseudo-regret -954. The code producing every reported number and figure is at `https://github.com/melikabaghi/state-exp3`. It pins the software environment used for the reported numbers, python 3.14.3 with numpy 2.4.2 and matplotlib 3.10.8, states the seed count for each study, lists one command per numbered study, and commits the stored outputs each figure reads so that every figure rebuilds without rerunning an experiment.

Does the drift bound describe the right scaling? The construction of Theorem 4 at $K = 40$, $T = 8000$, $h = d$ and 12 paired seeds, sweeping d from 10 to 200, the amplitude ε from 0.02 to 0.20 so that E_2 runs from 12 to 1193, and the drifting coordinates J from 1 to 16. We run four policies: uniform, action-level EXP3, STATE-EXP3, and the greedy policy on the exact stale vector m_t, the last because Theorem 4 claims to survive it. E_2 is measured on the realized instance rather than predicted, so the normalization uses the quantity the theorem uses.

Observed. The predicted scale $\sqrt{d\,E_2 \min\{1 + \log J, T/d\}}$ ranges from 38 to 377 across the grid and the measured regret tracks it (Figure 3). Normalized, the four policies read 0.230 to 0.429, 0.228 to 0.424, 0.238 to 0.417 and 0.237 to 0.437, a spread below 1.9 against a tenfold range in the predictor. The oracle-assisted policy has the largest mean ratio, 0.345 against 0.311, 0.314 and 0.310 for the three that never see m_t.

Interpretation. The three arguments enter the predicted scale differently and the collapse holds in all three sweeps, so it is the form of the bound rather than one fitted constant that tracks the data. That greedy play on the exact m_t sits at the top of the band rather than below it is the concrete content of the claim that the bound survives the oracle.

Limitation. Every cell has $T/d > 1 + \log J$, so only the $1 + \log J$ branch of the minimum is exercised and the T/d saturation branch, which needs d comparable to T, is not probed. A lower bound quantifies over all algorithms and four policies cannot establish that. What these runs check is the scaling of the construction.

Does Theorem 1 describe the right dependence on T, d and $\widehat{\overline{v}}$? A $4 \times 3 \times 5$ grid in row overlap, delay and horizon at $K = 40$, $|\mathcal{S}| = 8$ and five paired seeds, running the analyzed $m = d+1$ variant and reporting $R/\sqrt{(d+1)\widehat{\overline{v}}T \log K}$. A raw regret table cannot answer this and neither can a bound that holds with room to spare, since the question is about dependence rather than level.

Observed. Over the full grid the ratio spans 35.9 times, from 0.016 to 0.579. Dropping the near-degenerate column at $\widehat{\overline{v}} = 1.013$ leaves 7.1 times over 45 cells. At $\widehat{\overline{v}} = 3.656$ and $d = 10$ it moves only 1.35 times across a sixteenfold range of T, and at $d = 200$ it moves 2.57 times. Against rounds per interleaved copy $T/(d + 1)$ the three delays fall on one increasing curve (Figure 4).

Interpretation. The data are consistent with the predicted $\sqrt{T}$ dependence once each interleaved copy receives enough rounds, while much of the remaining variation is associated with the number

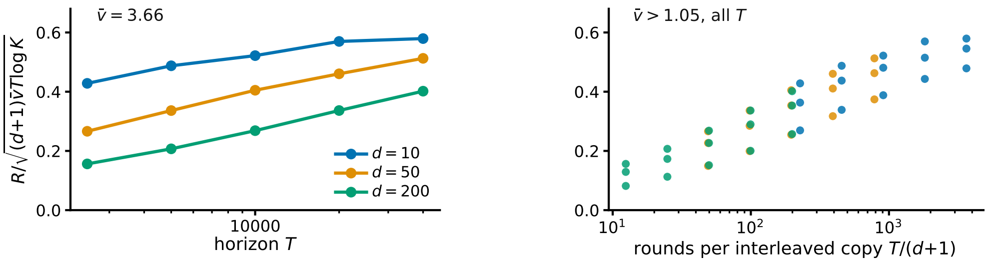

Figure 4: Study 9, the state channel. The plotted ratio would be flat if Theorem 1 captured the dependence on all three of T, d and $\widehat{\bar{v}}$. **Left,** horizon sweep at the largest effective dimension, one line per delay. **Right,** every cell with $\widehat{\bar{v}} > 1.05$ against rounds per interleaved copy, where the three delays fall on one curve.

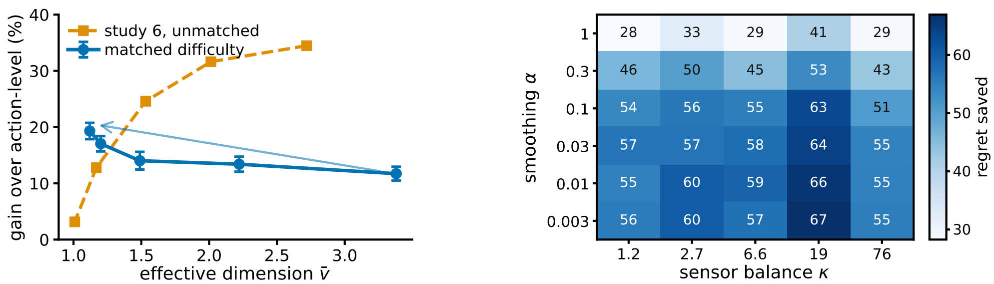

Figure 5: **Left,** study 10. The same overlap sweep with and without holding the task difficulty fixed. Matching the uniform-play regret reverses the trend, supporting the interpretation that the decline in Study 6 is driven by task-difficulty confounding rather than by the pooling mechanism itself. Bars are one standard error of the paired difference. **Right,** study 11, regret saved against action-level weighting over the smoothing pseudocount and the sensor balance, 10 paired seeds per cell.

of rounds available to each copy. At $d = 200$ and $T = 2500$ each of the 201 copies sees 12 rounds, so that cell is nowhere near the regime the bound describes. Together with the $m = 1$ comparison in Study 1, this suggests that much of the observed delay penalty comes from the interleaving device used in the analysis rather than from the pooled estimator itself.

Limitation. The degenerate column is degenerate for the reason study 6 gives, since at $\widehat{\bar{v}} \approx 1$ the rows of P agree, every action induces nearly the same loss, and both regrets collapse for a reason the bound cannot see. One natural explanation for the remaining looseness is ruled out rather than confirmed, since the realized $\sum_t v_t$ lies within 1.01 to 1.49 of the budget $\widehat{\bar{v}}T$, so the sup over play distributions is not the source.

Does pooling help because the rows overlap, or because overlap made the task easy? Study 6 sweeps the overlap and finds the advantage falling as $\widehat{\bar{v}}$ falls, which read at face value contradicts Theorem 1, whose bound improves as $\bar{v}$ falls. That sweep is confounded, since raising the overlap also flattens $c_t = P\theta_t$ across actions, collapsing the comparator gap so that both regrets vanish together. This study holds $K = 40$, $|\mathcal{S}| = 8$, $T = 10000$, $d = 50$ and the difficulty fixed, and varies only $\widehat{\bar{v}}$. Difficulty is the uniform-play regret $\sum_t \mathrm{mean}_a c_t(a) - \min_a \sum_t c_t(a)$, a property of the instance with no algorithm in it, and the amplitude of θ is bisected in every cell to hit a common target. The map keeps one contrast direction alive at every overlap and θ is aligned with it, which is what lets the gap survive as the rows agree.

Observed. Across 20 paired seeds the relative gain rises from 11.7 per cent at $\widehat{\bar{v}} = 3.380$ to 19.3 per cent at $\widehat{\bar{v}} = 1.120$, with absolute gains 85.5 ± 8.9 to 106.2 ± 7.9 and a correlation of -0.929 between $\widehat{\bar{v}}$ and the gain. Study 6's unmatched sweep falls from 34.5 to 3.1 per cent over a comparable range (Figure 5).

Interpretation. The two sweeps disagree in sign, and the matched-difficulty sweep agrees with the overlap dependence suggested by Theorem 1, although this experiment uses the practical $m = 1$ variant rather than the theorem's $m = d + 1$ algorithm. Overlap is what pooling exploits, and study 6 measures overlap confounded with an instance becoming trivial. This also explains study 9's degenerate column without appealing to it.

Limitation. At the two highest overlaps some seeds cannot reach the common target and the realized difficulty is 5 to 13 per cent below it, so the matching is close rather than exact. Since a smaller gap should if anything shrink the advantage, the measured rise is conservative in that direction. The matched gains are also smaller in magnitude than the unmatched ones, which is what fixing the difficulty costs.

Does the matched-difficulty trend survive a second configuration? Study 10 is a single design point, so the same procedure was re-run at $K = 20$, $|\mathcal{S}| = 12$, $T = 8000$, $d = 20$ and 20 paired seeds, with the common target lowered to 500 so that it is reachable at every overlap. Nothing else changes.

overlap	$\widehat{v}$	R_{unif}	state	action	paired gain
0.00	3.440	500.0	364.7	407.2	42.5 ± 8.2
0.35	2.225	500.0	346.8	401.3	54.6 ± 8.3
0.65	1.518	500.0	309.3	372.4	63.2 ± 6.4
0.85	1.240	493.6	271.8	346.0	74.3 ± 7.2
0.95	1.191	490.1	256.9	327.6	70.7 ± 7.8

Observed. The relative gain rises from 10.4 to 21.6 per cent as $\widehat{v}$ falls from 3.440 to 1.191, and the correlation between $\widehat{v}$ and the absolute gain is -0.972 against study 10's -0.929. Realized difficulty spans 490.1 to 500.0, a 1.020 times spread, so the matching is tighter here than in study 10.

Interpretation. The direction study 10 reports is not an artifact of its single configuration. Two independent design points, differing in K, $|\mathcal{S}|$, T and d, give the same sign and a comparable magnitude.

Limitation. Both configurations use the same construction and the same matching routine, so this tests configuration sensitivity and not construction sensitivity. Both run the practical $m = 1$ variant. An earlier attempt at this replication silently failed because `match_scale` binds its target as a default argument, so the bisection saturated and the difficulty was never matched; the run reported here passes the target explicitly.

Is the plug-in fragile in the smoothing constant, and does that depend on coverage? Study 3 reports one map where the plug-in loses at $\alpha = 1$ and recovers as α falls, which is an anecdote about a single instance. Here α is swept over 1 to 0.003 against the sensor balance κ of Theorem 10, controlled from 1.2 to 76 by placing mass ρ on one scarce state and drawing rows around that mean, at $K = 40$, $|\mathcal{S}| = 8$, $T = 8000$, $d = 50$ and 10 paired seeds.

Observed. No cell loses to action-level weighting, the gain running from 28.3 to 66.9 over the 30 cells. The gain roughly doubles as α falls from 1 to 0.03 and is flat below that, while moving along κ at fixed α changes it far less, the worst-to-best plug-in regret within a κ column being 1.07. At the better α the plug-in is indistinguishable from known P, the paired difference clearing two standard errors in one of the five columns and changing sign in another.

Interpretation. Of the two parameters the analysis names, the one that matters here is the one the analysis treats as free. A Laplace default costs about half the available gain at every coverage level tried, and coverage imbalance over a sixtyfold range costs much less than that. Taken with study 2 this is further evidence that the difficulty in the unknown-P setting is certification rather than estimation.

Limitation. This is one map family with κ varied through a single scarce state, so it does not rule out a construction in which κ is the binding parameter, and Theorem 10 still carries κ for a worst case this family need not contain. The flatness below $\alpha = 0.03$ says the leading behavior stops moving, not that any fixed α is safe at every horizon.

How far is the method from its certificate, and the certificate from useful? Study 2 reports these numbers in separate columns, leaving the reader to assemble them. Here action-level weighting,

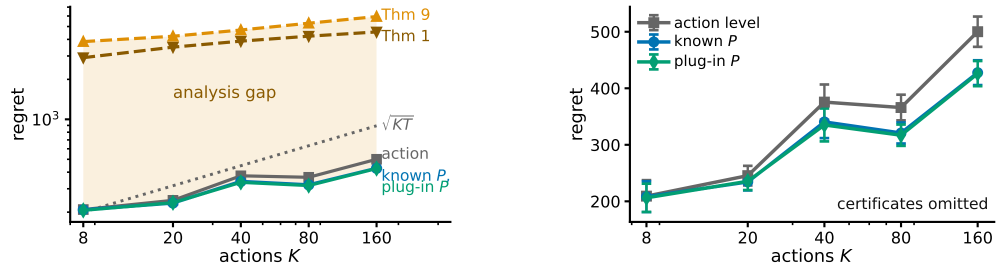

Figure 6: Study 12. **Left,** the three measured arms and the two certificates on one axis, $|\mathcal{S}| = 8$, $T = 5000$, $d = 50$, 12 paired seeds, every measured arm at $m = 1$. The shaded region is what the analysis costs rather than what the method does. **Right,** the same measured arms rescaled with the certificates removed, which is the only way to see that known P and the plug-in coincide. Bars are one standard error.

STATE-EXP3 with P given, and the same estimator with $\hat{P}_t$ updated every round are measured on the same paired seeds across K from 8 to 160, and drawn beside the two certificates. Theorem 1 is the certificate for known P and analyzes $m = d + 1$, while Theorem 10 is the warm-start rate and analyzes an algorithm that freezes $\hat{P}$ and restarts, so both are plotted to show the size of a gap and not as like for like.

Observed. The plug-in stays within 1.6 per cent of known P at every K, the two curves being visually indistinguishable, while action-level weighting costs 1.4 to 17.4 per cent more than the plug-in as K rises twentyfold. Theorem 1 sits 10.7 to 14.8 times above the measured plug-in and Theorem 10 sits 14.0 to 18.7 times above it. The warm-start certificate also stays above $\sqrt{KT}$ at every K tried, by factors of 19.2 down to 6.6.

Interpretation. The vertical gap reflects analytical conservatism rather than observed algorithmic performance. Estimating P online is close to free on this family, so the order of magnitude between the measured arms and either certificate is conservatism in the proof. That the warm-start rate never drops below $\sqrt{KT}$ here is the concrete form of the statement in Section 4 that it does not certify the channel it analyzes.

Limitation. The measured arms run $m = 1$ and Theorem 1 analyzes $m = d + 1$, and the plug-in is not the warm-start algorithm Theorem 10 analyzes, so neither vertical gap is a statement about a bound being loose for the algorithm it covers. A conservative bound and a well-behaved algorithm are consistent with a third possibility, that the family is easy.

Can the coarsening be chosen from data? Theorem 3 takes the representation as given and study 7 uses the latent cluster count, which no algorithm knows. This study supplies a rule and measures what it costs, with no theorem behind it. A prefix of 1200 rounds runs the identity representation and estimates $\hat{\theta}(s)$ by averaging the outcomes seen at each state. Sorting states by $\hat{\theta}$ and merging adjacent pairs greedily gives one candidate at every granularity from 16 down to 1, each scored by the quantity Theorem 3 bounds, $\sqrt{2(d+1)V(g)\log K} + 2\sum_t \hat{\delta}_t(g)$, and the argmin runs the remaining rounds. Against it are the identity, the true clusters, and the best candidate on the ladder scored in hindsight, over 24 paired seeds.

Observed. The rule fails. It selects 15 groups at heterogeneity 0 and 16 at 0.12, against a true cluster count of four, and lands at 912.9 ± 30.1 and 944.7 ± 31.1 where the identity gives 921.0 ± 30.5 and 952.7 ± 32.4, so it captures under one per cent of the available gain. The true clusters give 843.7 ± 27.9 and 883.0 ± 31.1, and hindsight over the same ladder gives 833.5 ± 27.8 and 841.5 ± 23.6.

Interpretation. Hindsight nearly matches the true clusters, so the ladder does contain a good representation and the failure is the criterion rather than the candidate set. Measuring $\hat{\delta}$ at the four-group representation locates it. At heterogeneity 0 the true within-group spread is exactly zero while the prefix estimates it at 0.154, which is sampling noise, and the bound multiplies that by $2(T - T_0)$ to give a phantom bias of 1483 against a real variance saving of 1261. The bias term is the conservative half of Theorem 3, and a noisy plug-in estimate of it is enough to forbid every merge.

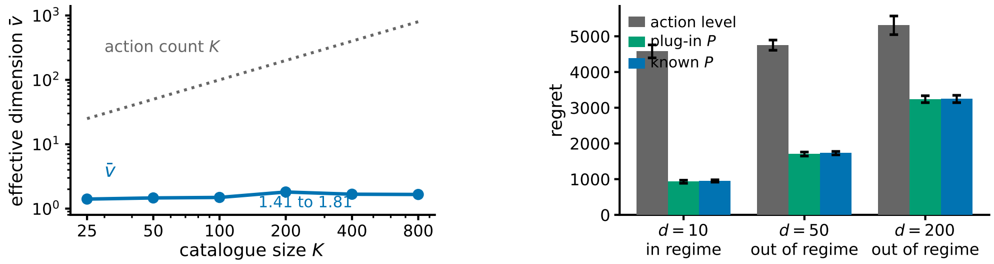

Figure 7: Study 14, a recommendation funnel. **Left,** the effective dimension stays near two while the catalogue grows thirty-twofold, so the gap between K and $\widehat{\bar{v}}$ comes from the catalogue having categories, not from a chosen K. **Right,** the three arms at $T = 20000$ and 8 paired seeds, labelled by whether the regime condition of Section 4 holds. Bars are one standard error. No real data is used anywhere in this study.

Limitation. One selector on one family, and a negative result about this rule is not a negative result about data-driven selection. The obvious repairs, debiasing $\hat{\delta}$ for its sampling noise or scoring candidates on held-out loss instead of on the bound, are not evaluated here. The prefix is also charged to every arm equally, so the comparison understates what a selector that spent less on estimation might do.

Is "many actions resolved by few states" a natural regime? Every study so far draws P from a Dirichlet, which is the assumption under test rather than evidence for it. This one builds an environment to the shape of a recommendation funnel instead. Actions are $K = 200$ catalogue items, states are $|\mathcal{S}| = 6$ ordered engagement depths, and the outcome is a delayed conversion. Three structural facts are imposed because funnels have them, namely that items fall into a modest number of categories sharing an engagement profile, that item quality is heavy-tailed, and that the loss falls with engagement depth and drifts seasonally rather than adversarially. The generated catalogue puts 74 per cent of its mass on no engagement and 1.4 per cent on the deepest.

Observed. The effective dimension is $\widehat{\bar{v}} = 1.97$ against $K = 200$, a hundredfold gap, and it stays between 1.41 and 1.81 as the catalogue grows from 25 to 800 items (Figure 7). Against action-level weighting the state-pooled variant saves 79.3, 63.6 and 38.9 per cent at $d = 10$, 50 and 200, the largest margins anywhere in this appendix. The plug-in matches known P at every delay, the paired difference never reaching two standard errors. The regime condition $\bar{v}(d+1)\log K \le K + d$ holds at $d = 10$ and fails at $d = 50$ and $d = 200$.

Interpretation. The separation between K and $\widehat{\bar{v}}$ arrives from catalogue structure rather than from a tuned parameter, and it does not close as the catalogue grows, which is the claim the Dirichlet families cannot make. That the condition Theorem 1 needs fails at the delays a funnel has, while the practical variant still saves most of the regret, is the same message as study 1 in the setting that motivates the paper. It is evidence for Conjecture 5 rather than for the theorem.

Limitation. *No real data is used.* There is no interaction log in this environment, nothing here is fitted to one, and this is not a semi-synthetic benchmark in the usual sense of that phrase. The structure is imposed by hand from qualitative properties, so it shows that the regime is describable and self-consistent, not that any deployed system occupies it. The arms run $m = 1$, which carries no guarantee, and the environment has no adversarial component, so it exercises the pooling channel and not the drift channel.

Which assumption produces which benefit? Comparing against action-level weighting alone leaves open whether the gain comes from the state structure or from an implementation detail. We run five policies on the same paired seeds over nine settings. Besides action-level weighting and STATE-EXP3 they are EXP3-IX (Neu, 2015), the implicit-exploration variant of EXP3, an importance weight; *naive sharing*, which credits $P(S_t \mid a)X_t/\bar{p}(S_t)$ using the unconditional state distribution and so isolates the value of the denominator $q_t = x_t^\top P$ from the value of sharing at all; and a *best-state rule*, which estimates $\hat{\theta}(s)$ online, takes the best state, and plays the action most likely to reach it. Two settings are added to the earlier families, one where the best action mixes

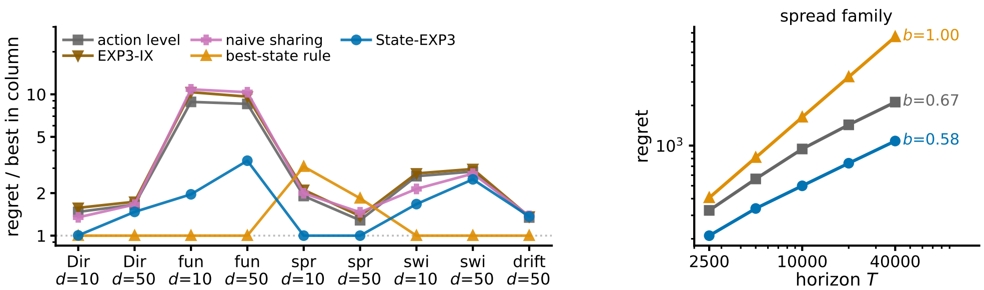

Figure 8: Study 15. **Left,** regret relative to the best arm in each family, so 1 is that column's winner, over five policies, nine settings and 10 paired seeds. The best-state rule leads seven columns and STATE-EXP3 leads two. **Right,** growth in T on the one family built so the rule's assumption is false, with fitted exponents. The left panel is a worst case over families chosen here; the right panel is why that is not a worst case over the model.

over several good states rather than maximizing the single best one, and one that is obliviously non-stationary. Running a bandit over states and mapping the values back through P is not a separate baseline, since $\sum_s P(s \mid a)\mathbf{1}\{S_t = s\}X_t/q_t(s)$ equals $P(S_t \mid a)X_t/q_t(S_t)$ identically, so that method *is* STATE-EXP3.

Observed. STATE-EXP3 does not win most columns. The best-state rule leads seven of the nine and STATE-EXP3 leads two, and over these nine settings the rule is also better on the median, 1.00 against 1.47 times the column winner, and on the worst case, 3.08 against 3.39 (Figure 8). EXP3-IX is 1.5 to 17.1 per cent worse than plain action-level weighting everywhere, and naive sharing ranges from 18.7 per cent better to 22.7 per cent worse than it. On the spread family the ordering inverts, STATE-EXP3 leading by 47.6 and 22.2 per cent while the best-state rule falls 61.5 and 43.0 per cent behind action-level weighting. Sweeping the horizon there, the rule's regret grows with fitted exponent 1.00 and STATE-EXP3's with 0.58.

Interpretation. The best-state rule buys its lead with an assumption, that the best action is the one most likely to reach the single best state. Where that holds it is very strong, and in the funnel it holds in 10 of 10 seeds because engagement depth is ordered, which is why it leads there by a factor of two. Where it fails the rule commits to a fixed action and pays regret linear in T, which no exploration rate repairs, and the exponent of 1.00 is that happening. The comparison therefore separates the assumptions rather than ranking the methods. STATE-EXP3 assumes nothing about the shape of θ. Across these nine selected settings its largest regret ratio to the column winner is 3.39, and the rule's suite maximum is slightly smaller at 3.08, but the rule's fitted growth on the spread family is linear. Action-level weighting also carries a sublinear guarantee, so what separates the arms here is the constant rather than the rate. STATE-EXP3's paired standard errors are the smallest of the five at 5.8 per cent of the mean against 23.6 for the rule. That naive sharing can be worse than no sharing is the separate point that the denominator, not the sharing, is what Proposition 7 buys.

Limitation. Nine settings chosen here are not a worst case over the model, and the median and worst-case columns above should be read as describing this set only. A practitioner whose problem has the ordered structure of the funnel should use the best-state rule, and this appendix does not identify when that structure can be verified in advance. On the drift family no arm's paired difference from action-level weighting reaches two standard errors, the largest being the best-state rule's $+24.6 \pm 12.4$, so that column is reported as an absence of signal rather than a result.

Does pooling still help against the strongest published delayed-bandit algorithm? Every comparison so far is against action-level weighting, EXP3-IX, naive sharing, or the best-state rule. None is the strongest published method for delayed feedback: that is the hybrid Tsallis-plus-entropy FTRL of Zimmert and Seldin (2020), whose delay term is additive rather than multiplicative. We run it on both headline families, with the same environments, seeds and delay convention, under three tunings.

family, cell	STATE-EXP3	Zimmert and Seldin	same, grid-best	STATE-EXP3, grid-best
funnel, $d = 10$	949.0	2959.4	2413.8	27.0
funnel, $d = 50$	1728.1	3557.0	2556.4	95.9
funnel, $d = 200$	3244.8	4734.7	2807.7	199.7
overlap 0.00	661.5	731.8	610.4	202.4
overlap 0.65	584.1	673.1	544.0	51.2
overlap 0.95	461.4	540.2	419.4	34.3

Observed. With STATE-EXP3 at the single tuning used everywhere else here and Zimmert and Seldin (2020) at the best of a six-point grid around its own value, STATE-EXP3 wins every cell, cutting regret by 31.5 to 67.9 per cent on the funnel and 9.6 to 14.6 per cent on the matched sweep. With both at their grid-best, over seven multipliers for Zimmert and Seldin (2020) and six for STATE-EXP3, STATE-EXP3 wins every cell again and by more, 66.8 to 98.9 per cent. With Zimmert and Seldin (2020) at its grid-best and STATE-EXP3 left at its paper tuning, the third column against the first, Zimmert and Seldin (2020) wins the funnel at $d = 200$ and all three matched cells, four of six.

Interpretation. No column is a symmetric protocol. The two grids are finite and unequal, and STATE-EXP3's grid-best sits at its top multiplier in every cell, so its own minimum may lie outside the range searched; the truncation runs against STATE-EXP3. Where it is not held to the worse tuning of the two, the state channel helps against the strongest published method, which is expected, since that method is action-level by construction and cannot read the state at all. Grid tuning helps STATE-EXP3 far more than it helps Zimmert and Seldin (2020), which is consistent with pooling cutting estimator variance and lower variance admitting a larger learning rate.

Limitation. The third column against the first is a real loss and is reported as one. These are our own synthetic families, and a method built for adversarial worst cases is not shown at its best on benign instances. Every grid-best multiplier is chosen by reading these benchmarks, so none of the last three columns is a procedure a practitioner could run; only the first is. Eight paired seeds on the funnel and ten on the matched sweep.

Taken together the sixteen studies cover both channels. Studies one, six, seven, nine, ten, fourteen and fifteen probe the state channel. They are consistent with the effective dimension, rather than $|\mathcal{S}|$ or K, governing what pooling buys, with the qualification that study ten had to control for task difficulty before study six's sweep pointed the way Theorem 1 does, and study nine shows where the theorem stops describing the dependence and attributes that to the interleaving. Studies two to five, eleven and twelve probe what happens when the action-to-state matrix must be estimated, and locate the difficulty in certifying the estimate rather than in forming it, study eleven adding that the smoothing constant matters more than the coverage imbalance the analysis carries and study twelve putting the measured arms and their certificates on one axis. Study thirteen closes the representation thread by showing that the bound of Theorem 3 is too conservative to serve as its own selection rule. Study fourteen is the only one whose action-to-state matrix is not drawn from a Dirichlet, and it reports the largest margins here, though on a structure imposed by hand rather than measured. Study fifteen widens the baseline set and finds STATE-EXP3 beaten in seven of nine settings by a heuristic that assumes the best action is the one reaching the single best state, which holds on several of these families and fails with linear regret when it does not. Study sixteen places STATE-EXP3 against the strongest published delayed-bandit method. It is ahead in every cell when both are tuned alike, and behind in four of the six when that method alone is grid-tuned. Study eight is the drift channel, and its collapse across d, E_2 and J is the strongest evidence here that Theorem 4 has the right form. No study runs a method that uses both channels, which is the gap Section 8 describes.

J Why the delay does not become additive

Run STATE-EXP3 with $m = 1$, so $x_t \propto \exp(-\eta L_t)$ with $L_t = \sum_{r \le t-d-1} \hat{c}_r$. Let $\tilde{x}_r$ be the iterate that has also absorbed the outstanding estimates $G_r = \sum_{s=r-d}^{r-1} \hat{c}_s \ge 0$. Two terms appear.

The delay term is $\mathbb{E}[\langle x_r - \tilde{x}_r, \hat{c}_r \rangle]$. Since $\tilde{x}_r(a) \ge x_r(a) e^{-\eta G_r(a)}$ and $1 - e^{-z} \le z$, it is at most $\eta \mathbb{E}[\sum_a x_r(a) G_r(a) \hat{c}_r(a)] = \eta \sum_{s=r-d}^{r-1} \mathbb{E}[\langle x_r, \hat{c}_s \rangle]$, and x_r depends only on rounds $r - d - 1$ and earlier, hence on rounds before s, so $\mathbb{E}[\langle x_r, \hat{c}_s \rangle] = \mathbb{E}[\langle x_r, c_s \rangle] \le 1$. This term is $\eta d T$ for any unbiased estimator, which is the additive form.

The quadratic term is where it stops. The telescoping scores $\sum_a x_{r+d}(a)\hat{c}_r(a)^2$, with x_{r+d} the iterate that has absorbed rounds up to $r-1$, while Proposition 7 bounds the play-weighted sum, the one against x_r, by v_r. Carrying out the substitution, $\mathbb{E}[\sum_a x_{r+d}(a)\hat{c}_r(a)^2 \mid \cdot] \le \sum_s q_{r+d}(s)/q_r(s) \le |\mathcal{S}|/Z_r$ for $Z_r = \mathbb{E}_{a\sim x_r}[e^{-\eta G_r(a)}]$, and $Z_r \ge e^{-\eta\langle x_r, G_r\rangle}$, so the swap costs an exponential moment of $\langle x_r, G_r\rangle$.

No bound on that moment holds uniformly over state distributions. For this estimator $\langle x_r, \hat{c}_s\rangle = q_r(S_s)X_s/q_s(S_s)$, so the quantity is governed by $1/q_s(S_s)$, the reciprocal probability of the state actually observed. Its mean is $\sum_s q_s(s)/q_s(s) = |\mathcal{S}|$, which is why the delay term is fine. For a fixed q of full support the exponential moment is finite, $1/q(S)$ being bounded, and what fails is uniformity. Already at $|\mathcal{S}| = 2$ with $q = (\varepsilon, 1-\varepsilon)$, $\mathbb{E}[e^{\eta/q(S)}] \ge \varepsilon e^{\eta/\varepsilon} \to \infty$ as $\varepsilon \to 0$ while the mean stays at 2, so no bound in $|\mathcal{S}|$ and η alone is available, and q_s is set by the algorithm's own past play. Measured at $|\mathcal{S}| = 4$, the mean stays at 4.0 while $\mathbb{E}[e^{0.05/q(S)}]$ runs 1.22, 1.43 and then 1.3×10^{21} as the smallest state probability falls through 0.25, 0.019 and 0.0009.

None of this is needed: Appendix K controls the same swap without any moment of $1/Z_r$, by bounding the state-marginal ratio directly. The rest of this section records why the route through $1/Z_r$ fails, since it is the natural first attempt.

Flooring q restores the moment and costs the product back. Mixing with the uniform action at rate γ gives $q_t(s) \ge \gamma\bar{p}(s)$, so $\eta\langle x_r, G_r\rangle \le 1$ needs $\gamma \asymp \eta d\kappa|\mathcal{S}|$ under the balanced-sensor condition $\bar{p}(s) \ge 1/(\kappa|\mathcal{S}|)$, and the mixing cost γT then reproduces $\widetilde{O}(\sqrt{d\kappa|\mathcal{S}|T\log K})$. So the obstruction is not an artifact of one device. Empirically it does not bite, $1/Z_r$ staying below 1.33 at the tuned η over $d \in \{10, 50, 200\}$, which is what suggested that the obstruction was an artifact of the device rather than of the problem, as Appendix K confirms.

K Additive delay without interleaving

Interleaving cannot give an additive bound. Copy i, which absorbs a share V_i of the budget $V = \sum_i V_i$, has regret $\sqrt{2V_i\log K}$, and Cauchy–Schwarz over the m copies gives $\sqrt{2mV\log K}$, tight when the V_i agree, so the $\sqrt{m}$ in Theorem 1 is not an artifact of the analysis. An additive bound has to come from a single copy.

Throughout, $L_t = \sum_{r\le t-d-1}\hat{c}_r$ and $x_t \propto e^{-\eta L_t}$, so $L_{t+1} - L_t = \hat{c}_{t-d}$, with $\hat{c}_r = 0$ for $r < 1$. The filtration $\mathcal{F}_t$ is the environment's, including outcomes not yet delivered, as in Proposition 7; under it x_t is $\mathcal{F}_{t-d-1}$-measurable, so x_{r+d} is $\mathcal{F}_{r-1}$-measurable.

The drift lemma. Appendix J priced the change of measure from x_r to x_{r+d} through $1/Z_r$. That is avoidable. The pooled estimate satisfies

$$\langle x_t, \hat{c}_t\rangle = \sum_a x_t(a)\frac{P(S_t \mid a)X_t}{q_t(S_t)} = X_t \le 1$$

identically, on every path, because the denominator is the state marginal of the same distribution that supplies the numerator.

Lemma 18. *If $\eta \le 1/(e(d+1))$ then $x_{t+1}(a) \le (1+1/d)\,x_t(a)$ for every t and every a, surely, and hence $x_{t+d}(a) \le e\,x_t(a)$.*

Proof. Strong induction on t. For $t \le d$ nothing is delivered and $x_{t+1} = x_t$. For $t > d$, assume the claim for every $s < t$ and compose it over the d steps from $t-d$ to t, giving $x_t(a) \le (1+1/d)^d x_{t-d}(a) \le e\,x_{t-d}(a)$. Contracting with the fixed nonnegative vector $\hat{c}_{t-d}$ and using the identity above, $\langle x_t, \hat{c}_{t-d}\rangle \le e\langle x_{t-d}, \hat{c}_{t-d}\rangle = eX_{t-d} \le e$. With $Z_t = \sum_b x_t(b)e^{-\eta\hat{c}_{t-d}(b)}$ and $e^{-z} \ge 1-z$, this gives $Z_t \ge 1 - \eta\langle x_t, \hat{c}_{t-d}\rangle \ge 1 - \eta e \ge d/(d+1)$, so $x_{t+1}(a) = x_t(a)e^{-\eta\hat{c}_{t-d}(a)}/Z_t \le x_t(a)/Z_t \le (1+1/d)x_t(a)$. □

What the induction bounds is $\langle x_t, \hat{c}_{t-d}\rangle = X_{t-d}\,q_t(S_{t-d})/q_{t-d}(S_{t-d})$, the state-marginal ratio Appendix J could not reach. Individual $\hat{c}_r(a)$ remain unbounded; only their x_t-weighted mass is controlled. Cesa-Bianchi et al. (2019) prove the action-level version under $\eta \le 1/(Ke(d+1))$ and Thune et al. (2019) remove the K, using that their estimate is carried by the single action played, so the weight cancels into a same-action ratio. A pooled estimate is spread over every action and does not collapse that way; the identity above replaces the cancellation.

Centring. Let $\zeta_t = \hat{c}_t - X_t\mathbf{1}$, with $\zeta_r = 0$ for $r < 1$. Exponential weights is unchanged by a shift that is constant across actions, so x_t is the same iterate and this is a change of analysis, not of algorithm (Eldowa et al., 2024). Keeping $\gamma(s) = \mathbb{E}[X_t^2 \mid S_t = s]$ inside the state sum,

$$\mathbb{E}\big[\langle x_t, \zeta_t^2\rangle \mid \mathcal{F}_{t-1}\big] = \sum_s \gamma(s)\Big[\frac{\sum_a x_t(a)P(s \mid a)^2}{q_t(s)} - q_t(s)\Big] \le v_t - 1,$$

each bracket being nonnegative by Jensen and $\gamma(s) \le 1$, the final step being Remark 6. Bounding $\langle x_t, \hat{c}_t^2\rangle$ by v_t first and then subtracting does not work: it leaves $v_t - \mathbb{E}[X_t^2]$.

Proof of Theorem 2. Since $\zeta_r(a) \ge -X_r \ge -1$ we have $\eta\zeta_r(a) \ge -\eta$, and Lagrange's remainder gives $e^{-z} \le 1 - z + (e^\eta/2)z^2$ for $z \ge -\eta$. With $\log(1+y) \le y$, the potential $W_t = \sum_a \exp(-\eta\sum_{r\le t-d-1}\zeta_r(a))$ obeys $\log(W_{t+1}/W_t) \le -\eta\langle x_t, \zeta_{t-d}\rangle + (\eta^2 e^\eta/2)\langle x_t, \zeta_{t-d}^2\rangle$. Telescoping from $W_1 = K$ and using $\log W_{T+1} \ge -\eta\sum_{r\le T-d}\zeta_r(a^*)$ for the comparator a^*,

$$\sum_{r=1}^{T-d}\langle x_{r+d}, \zeta_r\rangle \le \sum_{r=1}^{T-d}\zeta_r(a^*) + \frac{\log K}{\eta} + \frac{\eta e^\eta}{2}\sum_{r=1}^{T-d}\langle x_{r+d}, \zeta_r^2\rangle.$$

The shift cancels between the two sides. As x_r is $\mathcal{F}_{r-d-1}$-measurable, $\mathbb{E}[\langle x_r, \hat{c}_r\rangle] = \mathbb{E}[\langle x_r, c_r\rangle]$ and $\mathbb{E}[\hat{c}_r(a^*)] = c_r(a^*)$, and the d rounds undelivered by $T+1$ contribute at most d, so

$$\mathbb{E}[R_T] \le \frac{\log K}{\eta} + \frac{\eta e^\eta}{2}\sum_r \mathbb{E}\big[\langle x_{r+d}, \zeta_r^2\rangle\big] + \sum_r \mathbb{E}\big[\langle x_r - x_{r+d}, \hat{c}_r\rangle\big] + d.$$

Lemma 18 gives $x_{r+d} \le e\,x_r$ surely, and x_{r+d} is $\mathcal{F}_{r-1}$-measurable, so the second sum is at most $e\sum_r \mathbb{E}[v_r - 1] \le eV^-$.

For the third, write $x_{r+d}(a) = x_r(a)e^{-\eta G_r(a)}/Z'$ with $G_r = \sum_{u=r-d}^{r-1}\hat{c}_u \ge 0$ and normaliser $Z' \le 1$, distinct from the one-step Z_t of the Lemma. Then $x_r(a) - x_{r+d}(a) \le x_r(a)(1 - e^{-\eta G_r(a)}) \le \eta x_r(a)G_r(a)$ for every a, trivially where the left side is negative. Since $\hat{c}_r \ge 0$ and x_r, G_r are $\mathcal{F}_{r-1}$-measurable, the conditional expectation is at most $\eta\langle x_r, G_r\rangle$, and $\mathbb{E}[\langle x_r, G_r\rangle] = \sum_{u=r-d}^{r-1}\mathbb{E}[\langle x_r, \hat{c}_u\rangle] \le d$: for each such u, x_r is $\mathcal{F}_{r-d-1}$-measurable and $r-d-1 \le u-1$, so it leaves the conditional expectation and $\mathbb{E}[\langle x_r, \hat{c}_u\rangle] = \langle x_r, c_u\rangle \le 1$. This is measurability, not independence, and at $u = r-d$ the two σ-fields coincide, so the step rests on the timing convention of Section 3. The sum is at most ηdT.

Finally $\eta \le 1/(2e)$ gives $e^{1+\eta} < 3.3$, so $\mathbb{E}[R_T] \le \log K/\eta + \frac{\eta}{2}(3.3V^- + 2dT) + d$. For $f(\eta) = A/\eta + B\eta$ restricted to $\eta \le \eta_{\max}$ one has $\min f \le 2\sqrt{AB} + A/\eta_{\max}$, in the interior and in the binding case alike; with $A = \log K$, $B = (3.3V^- + 2dT)/2$ and $\eta_{\max} = 1/(e(d+1))$ this is the stated bound. □

The cap on η binds only when $d \gtrsim T/(e^2\log K)$, and not at the settings of Appendix I: the funnel study's own rates are 0.0058, 0.0031 and 0.0016 at $d = 10, 50, 200$ against caps 0.0334, 0.0072 and 0.0018, so Theorem 2 covers those runs as they were executed. Over three seeds and 20000 rounds the induction has room to spare, $\max_a x_{t+1}(a)/x_t(a)$ reaching 1.006, 1.003 and 1.002 against the permitted 1.100, 1.020 and 1.005.

The two bounds are numerically close on the funnel, where $\bar{v} \approx 2$: the additive form carries $3.3(\bar{v} - 1) + 2d$ and the interleaved one $(d+1)\bar{v}$, and these cross near $\bar{v} = 2$. The additive form gains as $\bar{v}$ rises toward $|\mathcal{S}|$, and it describes the algorithm that is actually run, which is the reason to prefer it.

L Stationary sensors

Observation 19. The multiplicative term $\sqrt{\min\{|\mathcal{S}|, d\}T}$ is not forced by the known construction, and is absent at one endpoint of the present model class.

The $\Omega(\sqrt{\min\{|\mathcal{S}|, d\}T})$ construction of Esposito et al. (2023, Thm. 5.2) re-splits states across actions adversarially per block, so its action-to-state map is time-varying and lies outside the model class of Section 3. At $E_2 = 0$ with stationary P the problem is a stochastic delayed bandit, where delay enters additively (Joulani et al., 2013). One route to it is therefore unavailable and the term is absent at one endpoint, but neither a matching upper bound for all $E_2 > 0$ nor a lower bound ruling it out is established here.